# Lambertian Models of Reflectance and Rendering

## Lecture 17
## Notes for Probabilistic Models of the Visual Cortex

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# What this lecture covers

Light and the Lambertian model

Photometric stereo, bas-relief

Shape from shading

Harmonic analysis of lighting

Rendering

Summary

# Colors used in the figures

These notes follow Yuille's notes on Lambertian models of reflectance (Yuille, 2020), abbreviated G; Basri's two lectures on the Lambertian model (Basri, B1; Basri, B2); and Wang and Yuille's slides on rendering (Wang and Yuille, AN). A caption citing "B1 p. *N*" took its picture from page *N* of B1, and likewise for B2 and AN.

In the diagrams drawn for these notes and the slides, colors keep one meaning.

- Orange marks light sources (a sun and arrow at the image border give the light's direction in the image plane), the transformed or shadowed version of a quantity, and cast shadows.
- Blue marks the linear model and attached shadows.
- Red marks the circle of possible normals, the limits for a distant light, and the gradient of a loss.
- Teal marks the coefficients of the Lambertian kernel.

A normal map shows each normal as a color, as on the small key sphere beside it. In pictures taken from the sources, the captions say what the colors mark.

# Lighting and pose change the appearance of objects

Vision has to recover the shape and the material of objects from images, and an image depends on much more than the object: on the light sources (their number, directions and strengths), on the pose of the object relative to the camera, and on how its surface reflects light.

The same object can therefore produce very different images. Lighting changes the shading (how bright each point of a smooth surface is), the shadows and the highlights (the bright spots where a shiny surface reflects a light source toward the camera). Pose changes which parts of the object are seen at all.

For faces this matters in practice: Adini, Moses and Ullman (1997) found that two images of one person under different lighting can differ more than images of two different people under the same lighting, so a recognizer that compares images directly fails. Understanding the effect of lighting on images is challenging, but it can lead to a better interpretation of images, and it is the subject of this lecture.

# Lighting and pose change the appearance of objects

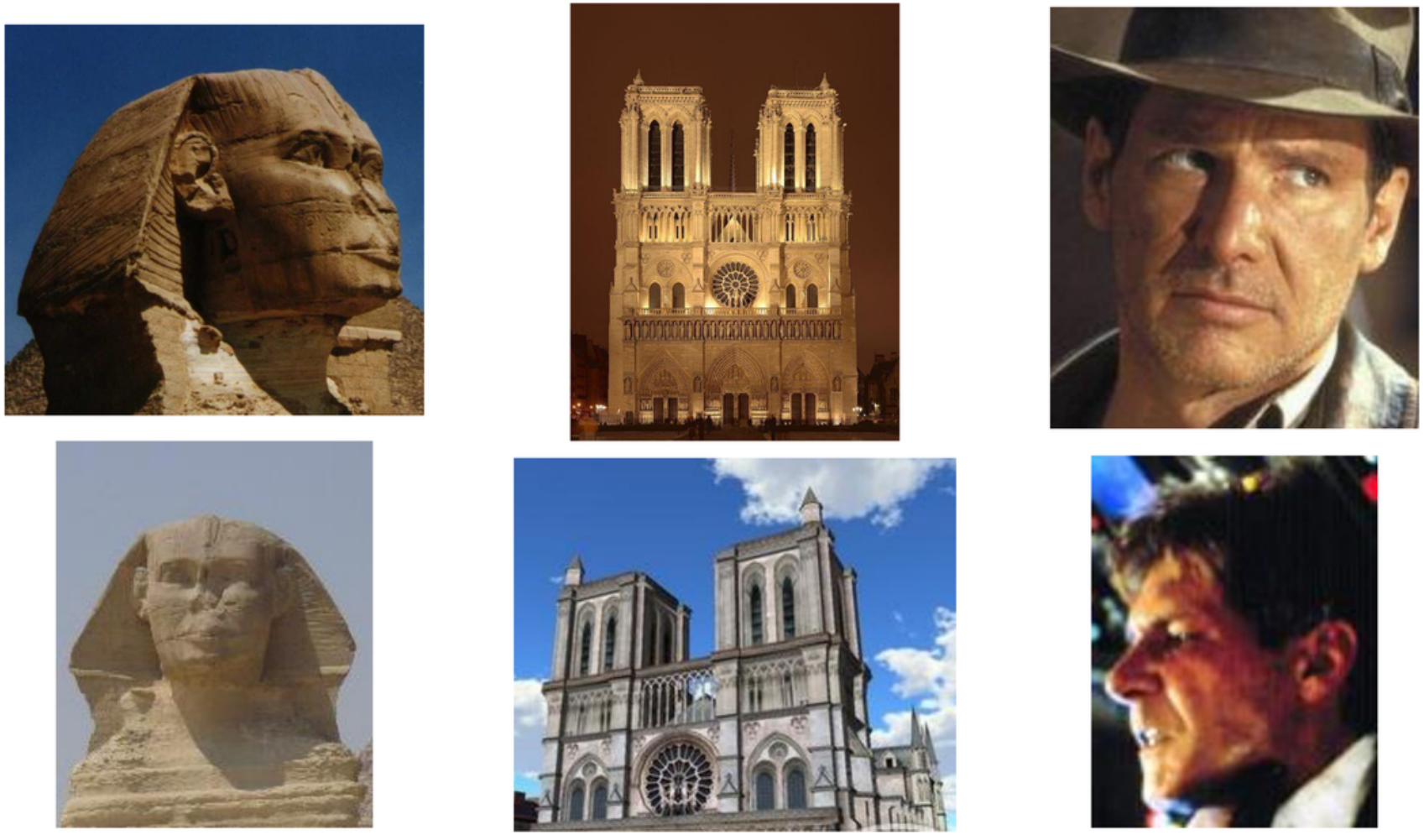


Figure 1: Each column shows one object in two images: the Sphinx (top: in profile, in sunlight; bottom: from the front, in haze), Notre-Dame (by night and by day) and an actor's face (two lightings and poses). B1 p. 2.

# Shading reveals the shape of smooth surfaces

Edges are the places where the intensity changes abruptly. They capture the outlines and the markings of objects, such as the windows and the wheels of a car, often enough to recognize it. On a smooth surface, such as a face, the intensity changes gradually, edges miss most of the shape, and the shape is carried by the shading. The relief on a sculpture may produce no edge at all: only its shading shows it.

How bright a point looks depends on three things: the shape of the surface (its normal at that point), how the surface reflects light, and the lighting. Inferring shape from shading requires a model of all three, and the Lambertian model of this lecture is the simplest one.

# Shading reveals the shape of smooth surfaces

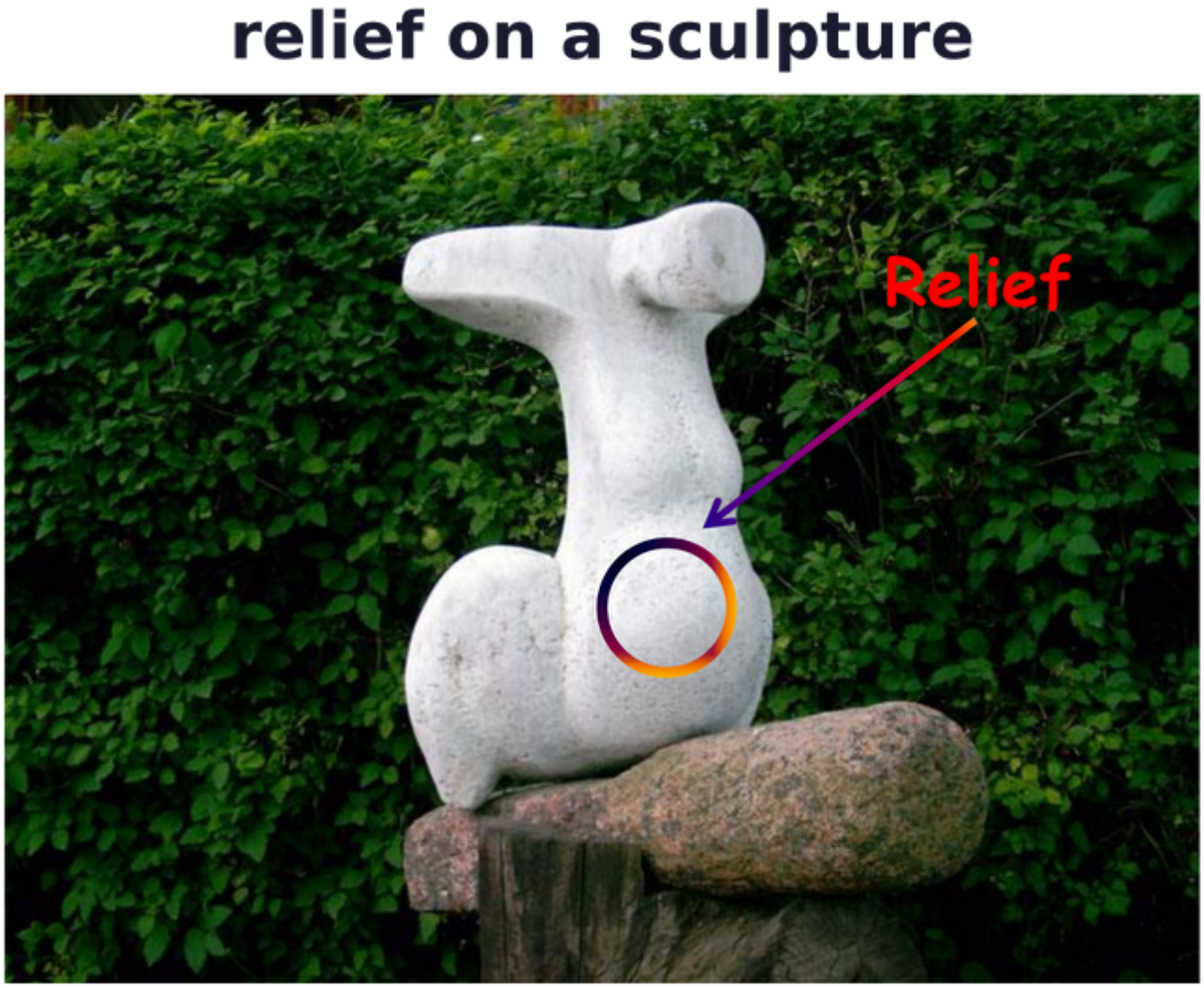



Figure 2: A sculpture whose relief (circled; the label is Basri's) shows only through its shading. Sculpture by Dumitru Verdianu; B1 p. 3.

# Edges miss the shape of smooth surfaces

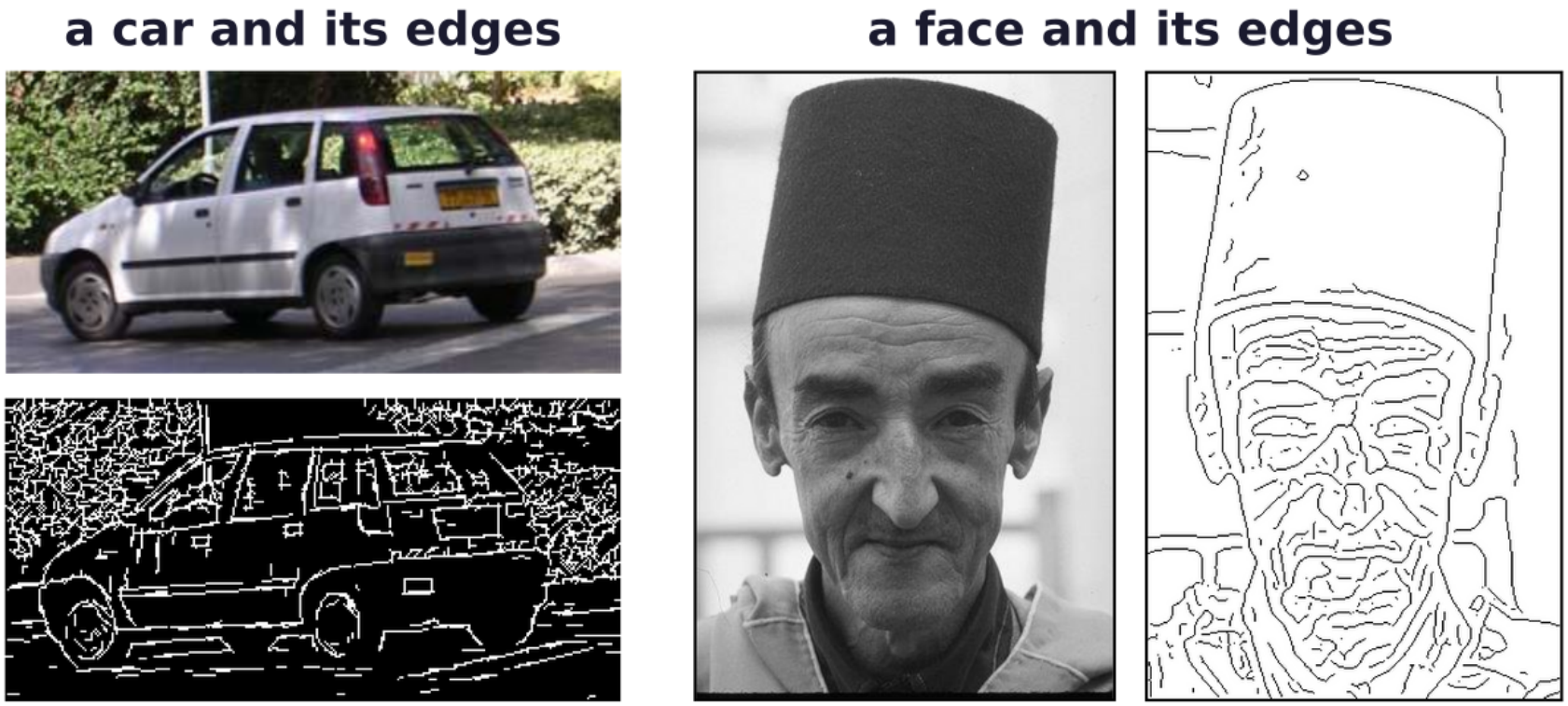



Figure 3: A car and a face with their edge maps (lines: the pixels where the intensity gradient is large). The edges give the car's outline, windows and wheels; for the face they give the outline, the eyes, the hat and the wrinkles, and almost nothing of the shape of the cheeks, the nose or the forehead. B1 pp. 4–5.

# Smooth surfaces reflect like mirrors, rough ones diffusely

Rays of light travel from the sources to objects, where they are absorbed or reflected, losing energy at each bounce. A camera captures the rays that pass through its focal center, the point where all the rays that form the image meet. We assume that the camera's response is linear: the value of a pixel is proportional to the light that reaches it.

How a surface reflects depends on its fine-scale roughness. A smooth surface reflects light in the mirror direction: with $\hat{n}$ the unit normal and $\hat{l}$ the unit vector toward the light, the reflected ray points along

$$\hat{r} = 2(\hat{n} \cdot \hat{l})\,\hat{n} - \hat{l},$$

at the same angle to the normal as $\hat{l}$, on the other side of it. A mirror is therefore seen only from that one direction. On a slightly rough surface the reflected light falls off around this specular (mirror) direction, and a very rough surface may reflect light equally in all directions (diffuse reflection), because its small facets face in many directions. On a glossy object the highlights are small bright patches where the mirror direction points toward the camera.

# Smooth surfaces reflect like mirrors, rough ones diffusely

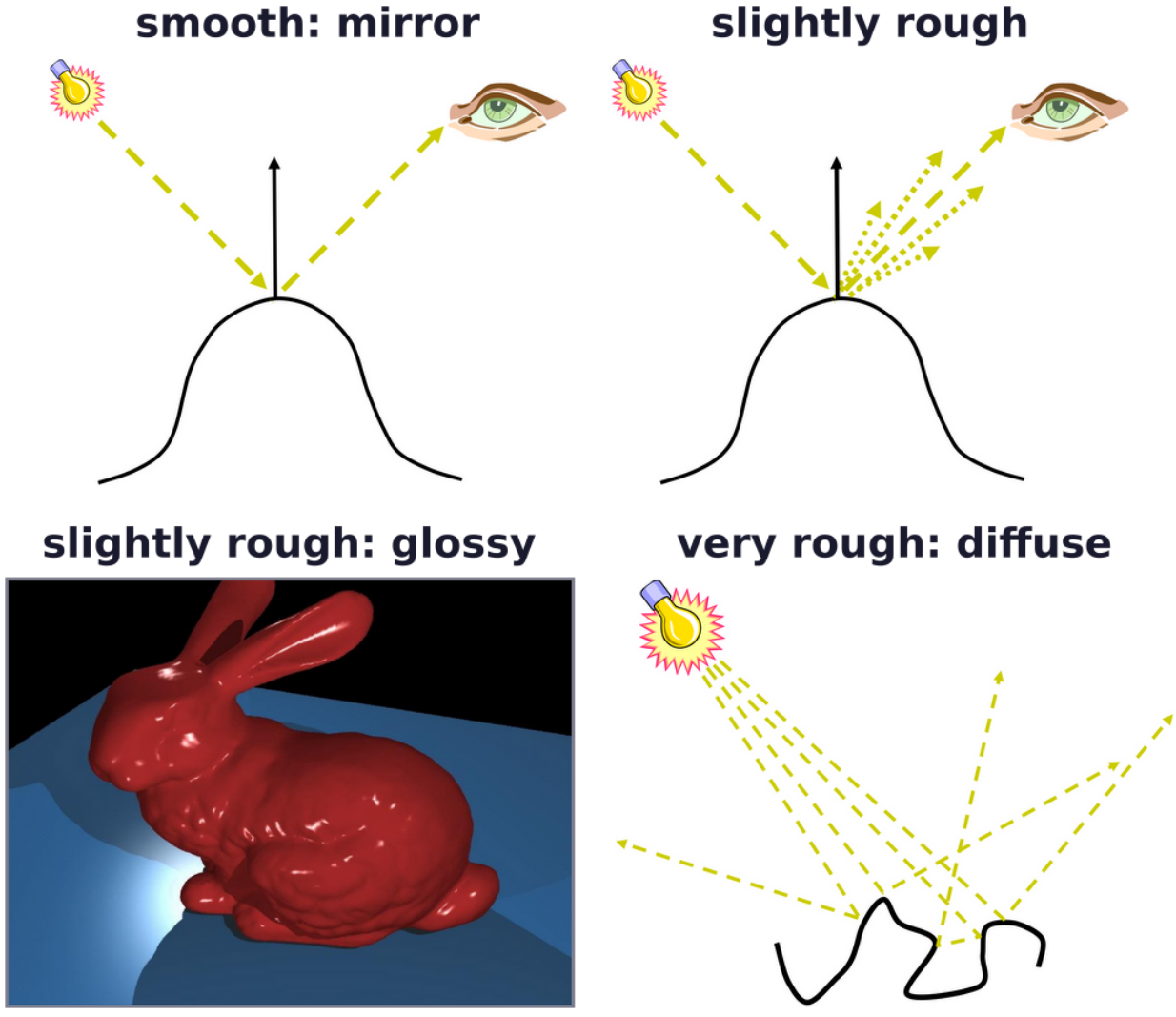



Figure 4: Top: a smooth surface (one reflected ray, at the angle of the incoming one) and a slightly rough one (dotted: light around the mirror direction); the eye marks the viewer. Bottom: a glossy rendered bunny, and a very rough surface. B1 pp. 9–12.

# The BRDF gives the light reflected toward each direction

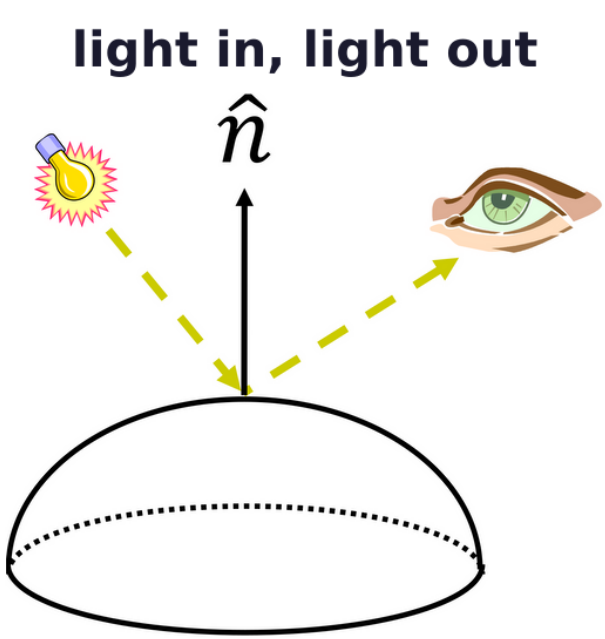
The Bidirectional Reflectance Distribution Function (BRDF) specifies how a surface reflects light. Directions are given in spherical angles about the normal: $\theta$ is the angle to the normal and $\phi$ the azimuth around it. For a unit of incoming light from the direction $(\theta_{in}, \phi_{in})$, the BRDF

$$f(\theta_{in}, \phi_{in}; \theta_{out}, \phi_{out})$$

is the light reflected toward $(\theta_{out}, \phi_{out})$. A mirror has a BRDF concentrated on the mirror direction; a Lambertian surface has a constant BRDF.

The BRDF can be expanded in a series of spherical harmonics, the analogue of a Fourier series for functions on the sphere. The Lambertian is the first term, and it is the only one for which the image is independent of the viewpoint. The higher terms give more realistic reflectances, used in computer graphics, but they make the image depend on where the camera is. A ploughed field shows such a dependence: it looks bright with the light from the front (behind the viewer), and dark with the light from the back, because its furrows then show the viewer their shadowed sides.

# The BRDF gives the light reflected toward each direction






Light from front Light from back

Figure 5: A BRDF: light in (from the bulb) and out (to the eye), about the normal $\hat{n}$; a ploughed field with the light from the front and from the back. B1 pp. 13–14.

# Lambertian surfaces follow the cosine law

A Lambertian surface reflects equally in all directions, so it looks equally bright from every viewpoint. Its brightness depends only on how much light reaches it. A beam of cross-section $A$ that meets the surface at angle $\theta$ to the normal covers the area $A/\cos\theta$, so each point receives $\cos\theta$ times the light it would receive from the same beam along the normal. This gives Lambert's law (Lambert, 1760),

$$I = E\,\rho\cos\theta \quad (\theta \le 90^\circ),$$

where $E$ is the strength of the light and $\rho$ the albedo, the fraction of the incoming light that the surface reflects (between 0 for black and 1 for white). For $\theta > 90^\circ$ the light is behind the surface and $I = 0$.

In vector form the light is $\vec{l} = E\,\hat{l}$ and the scaled normal is $\vec{n} = \rho\,\hat{n}$, with $\hat{l}$ and $\hat{n}$ unit vectors; since $\hat{l}\cdot\hat{n} = \cos\theta$, Lambert's law reads $I = \vec{l}^{\,T}\vec{n}$.

# Lambertian surfaces follow the cosine law

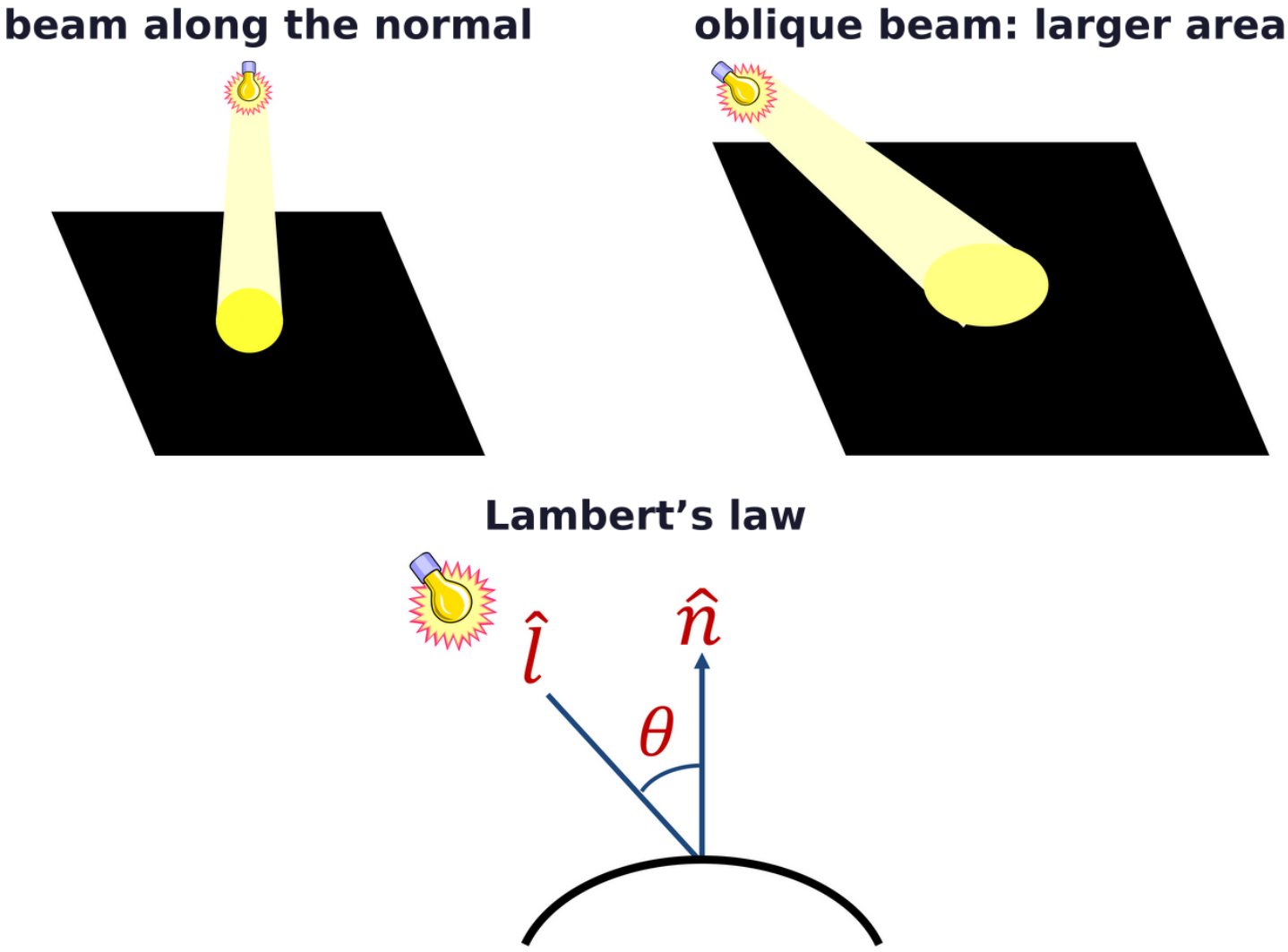



Figure 6: Top: the same beam along the normal and oblique; the oblique beam covers a larger area (yellow patch), so each point gets less light. Bottom: $\theta$ is the angle between the direction to the light $\hat{l}$ and the normal $\hat{n}$. B1 pp. 15–17.

# The Lambertian model is the simplest model of imaging

G writes the Lambertian model of an image as

$$I(\vec{x}) = a(\vec{x})\, \vec{n}(\vec{x}) \cdot \vec{s},$$

where $\vec{x}$ is a pixel, $I(\vec{x})$ the image, $a(\vec{x})$ the albedo, $\vec{n}(\vec{x})$ the unit surface normal, and $\vec{s}$ the light: $|\vec{s}|$ is its strength and $\vec{s}/|\vec{s}|$ its direction. This is Lambert's law with $a = \rho$ and $\vec{s} = \vec{l}$, written per pixel. It is also called the cosine rule: the image depends on the cosine between the normal and the light direction. It ignores specularity and inter-reflection (light that reaches a point after bouncing off other parts of the scene), but it is a good approximation much of the time and easy to analyze; BRDF models are much harder.

# The Lambertian model is the simplest model of imaging

The computed figures of these notes use one synthetic face: a depth map $z(x, y)$, an elliptical dome on a flat background with bumps and dents for the eyes, brows, nose, cheeks, lips and chin, seen from above along $z$. Its albedo is 0.86 on the face, darker at the brows and the lips, and 0.62 on the background.

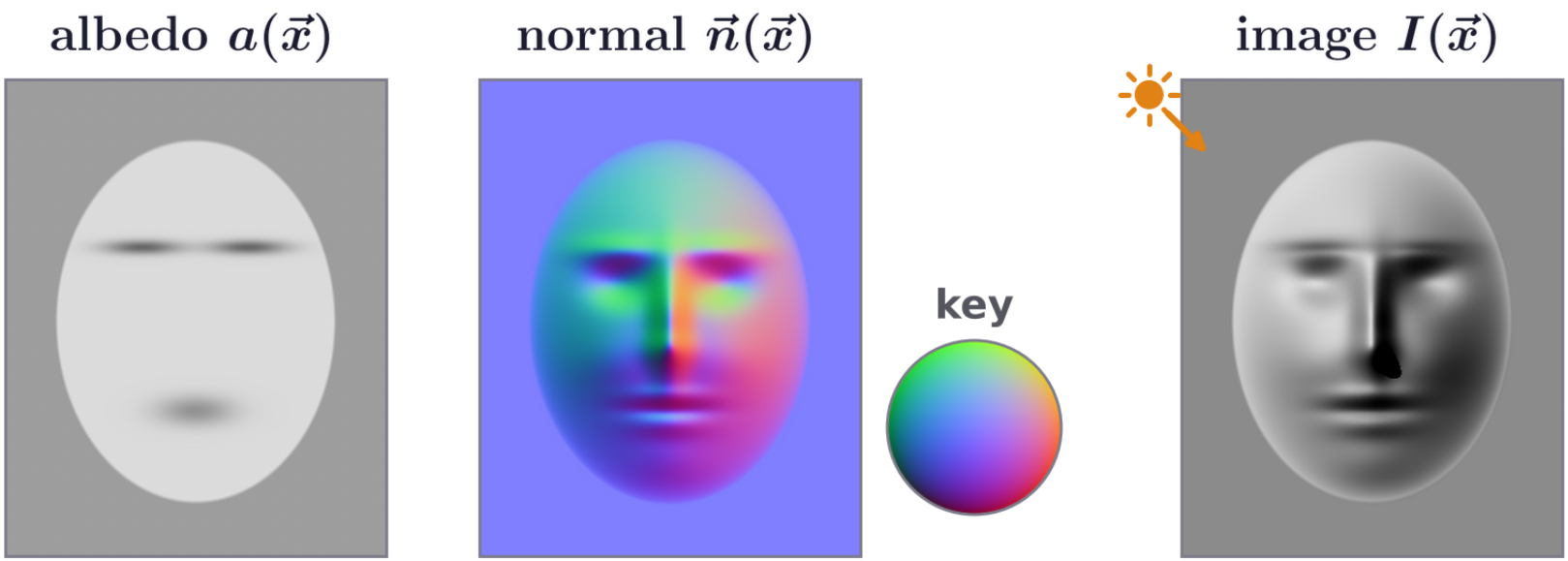



Figure 7: The synthetic face: albedo, normal (colored by direction as on the key sphere) and the image under one light (from the upper left, 62° above the image plane), with shadows.

# Without shadows, the image is linear in the light source

If $I_1(\vec{x}) = a(\vec{x})\vec{n}(\vec{x}) \cdot \vec{s}_1$ and $I_2(\vec{x}) = a(\vec{x})\vec{n}(\vec{x}) \cdot \vec{s}_2$ are the images under two lights, the image under both is $a(\vec{x})\vec{n}(\vec{x}) \cdot (\vec{s}_1 + \vec{s}_2) = I_1(\vec{x}) + I_2(\vec{x})$: images add, and scaling a light scales the image. This holds as long as no point turns away from a light; in the figure both lights come from 18° off the viewing direction and no point of the face turns away from either. Most importantly, the image does not depend directly on the viewpoint. A mirror is different: its image depends on the viewpoint, the light direction and its orientation.

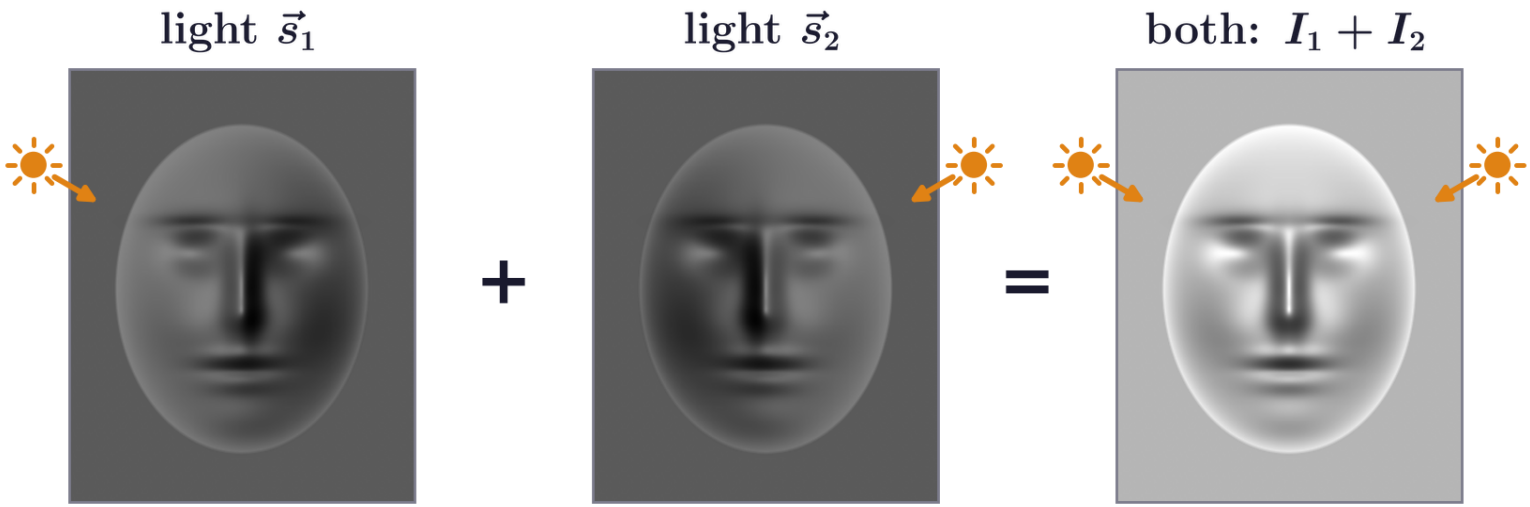



Figure 8: The synthetic face under the light $\vec{s}_1$ (upper left), under $\vec{s}_2$ (upper right), and under both; the third image is the sum of the first two.

# Convex lit from above looks like concave lit from below

There is a well-known perceptual ambiguity: a convex object lit from above cannot be distinguished from a concave object lit from below (Gregory, 1997). Humans resolve it by tending to perceive objects as convex, which is why the craters can look like bumps when the picture is turned upside down. An inverted (hollow) face mask even appears to be a real face, and this remains true with other cues, such as binocular stereo or motion, present.

The Lambertian model explains the ambiguity. Write the surface as a depth map $z(x, y)$, with the normal $\vec{n} \propto (-z_x, -z_y, 1)$, where $z_x = \partial z/\partial x$ and $z_y = \partial z/\partial y$ are the slopes. Replacing $z$ by $-z$ flips the first two components of the normal; flipping the first two components of the light, $\vec{s} = (s_1, s_2, s_3) \mapsto \vec{s}\,' = (-s_1, -s_2, s_3)$, leaves every product unchanged:

$$\vec{n}(-z) \cdot \vec{s}\,' = \vec{n}(z) \cdot \vec{s}.$$

So the dip $-z$ lit from the opposite side gives exactly the image of the bump $z$.

# Convex lit from above looks like concave lit from below

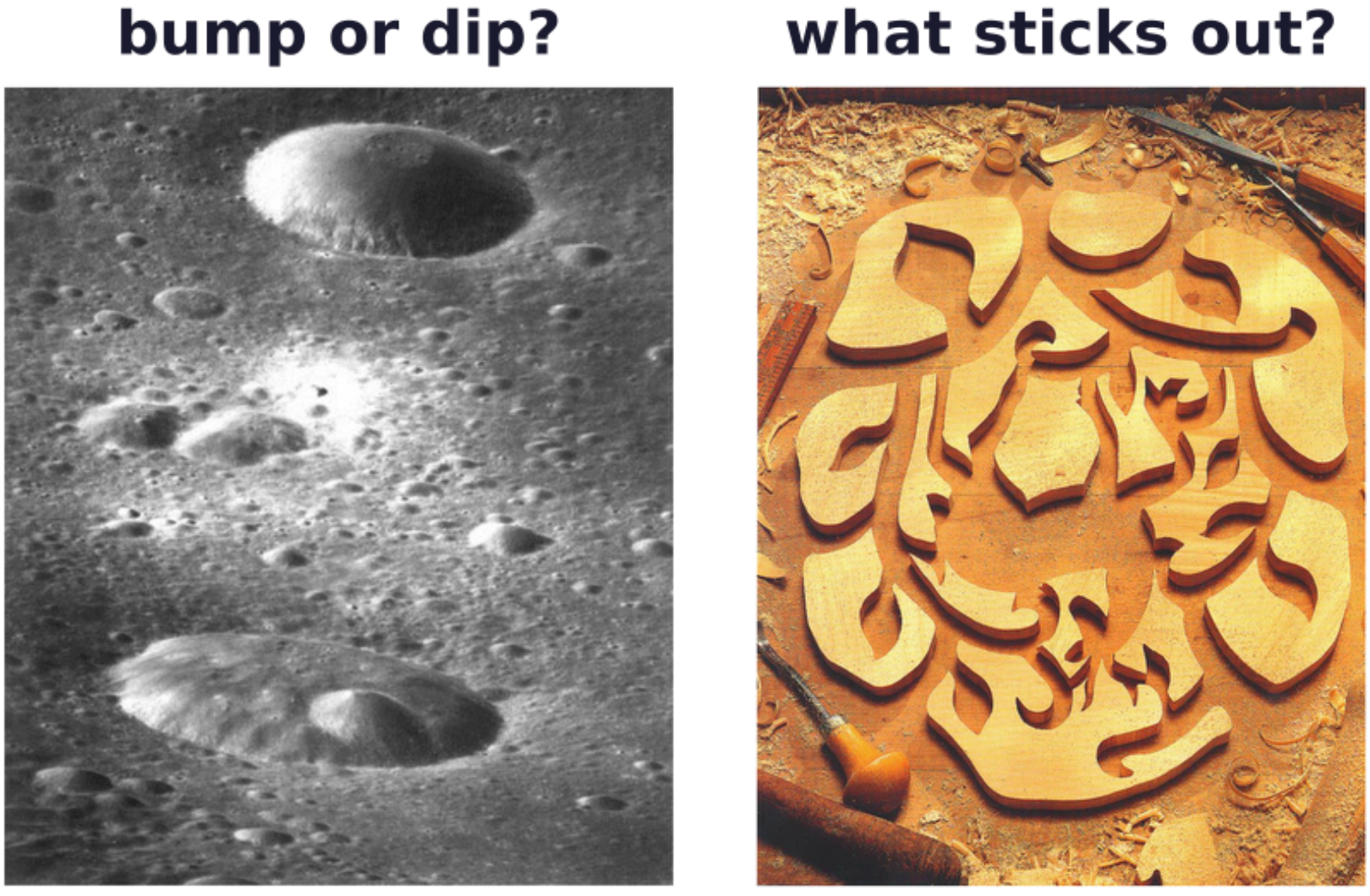



Figure 9: Bump or dip? Craters on the Moon, and a wood carving: which parts stick out? B1 pp. 6–7.

# A bump lit from above equals a dip lit from below

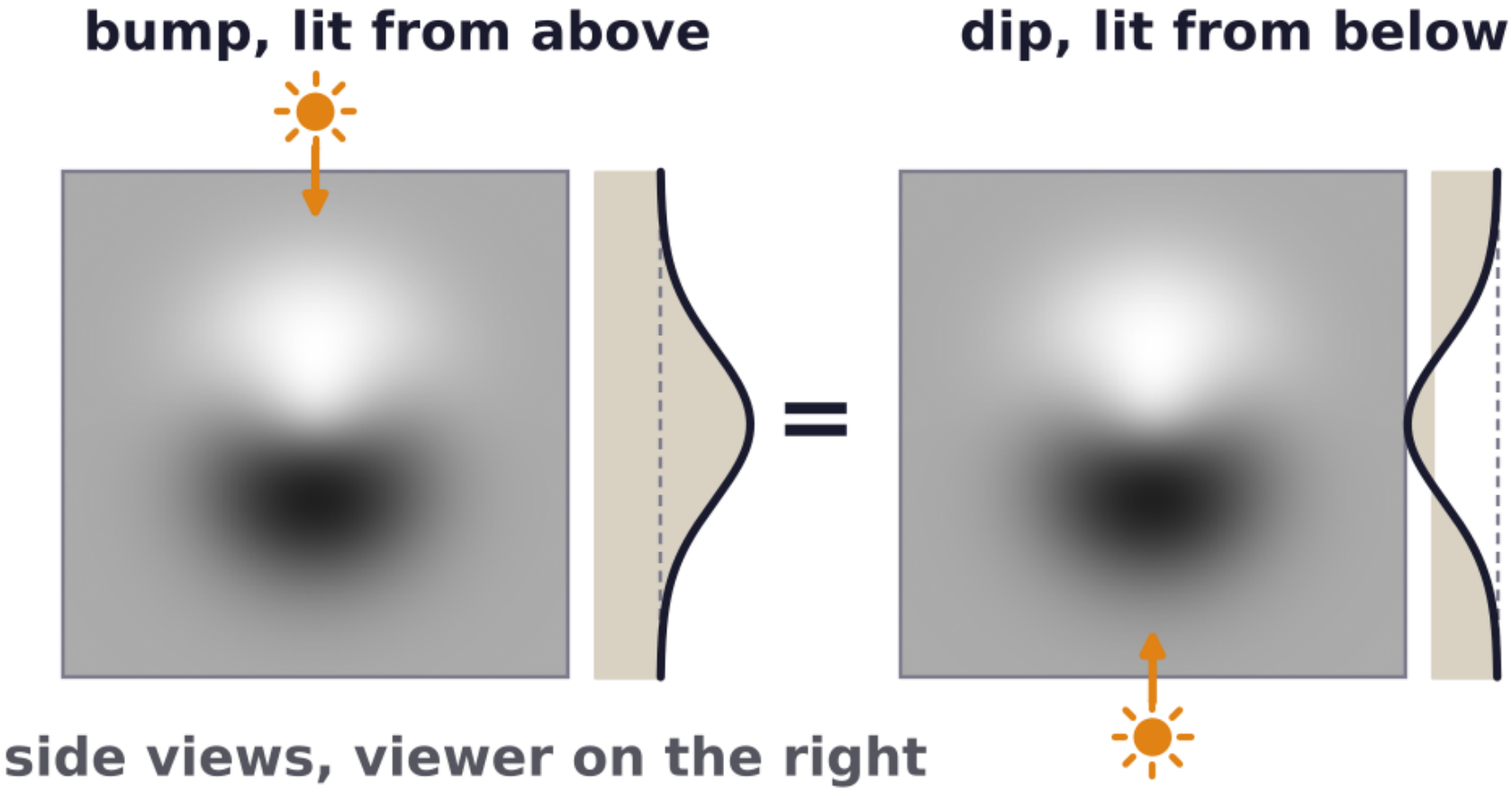



Figure 10: A Gaussian bump of height 0.32 lit from the top of the image (the light 40° above the image plane), and the dip $-z$ lit from the bottom. Beside each image its side view, viewed from the right (dashed: the flat background). The two images are identical.

# Attached and cast shadows break the linear model

The linear model must be modified to account for shadows, of two types. An attached shadow is a point whose surface faces away from the light, $a\,\vec{n}\cdot\vec{s} < 0$; the model would give a negative intensity, and the true one is 0. A cast shadow is a point that faces the light but is occluded from it by another part of the object. Attached shadows are handled by clipping each light's contribution at zero,

$$I(\vec{x}) = \sum_{\mu} \max\{a(\vec{x})\,\vec{n}(\vec{x})\cdot\vec{s}_{\mu},\, 0\},$$

where $\mu$ indexes the light sources. This is no longer linear in the light sources, which makes the analysis harder but still possible (G does it with spherical harmonics).

Cast shadows depend on the whole surface. A point at the top of a mountain and a point at the bottom of a mineshaft have the same normal (pointing up) and may have the same albedo, so the Lambertian model gives them the same intensity. Yet the bottom of the shaft is much darker: only the sky straight overhead reaches it.

# Attached and cast shadows break the linear model

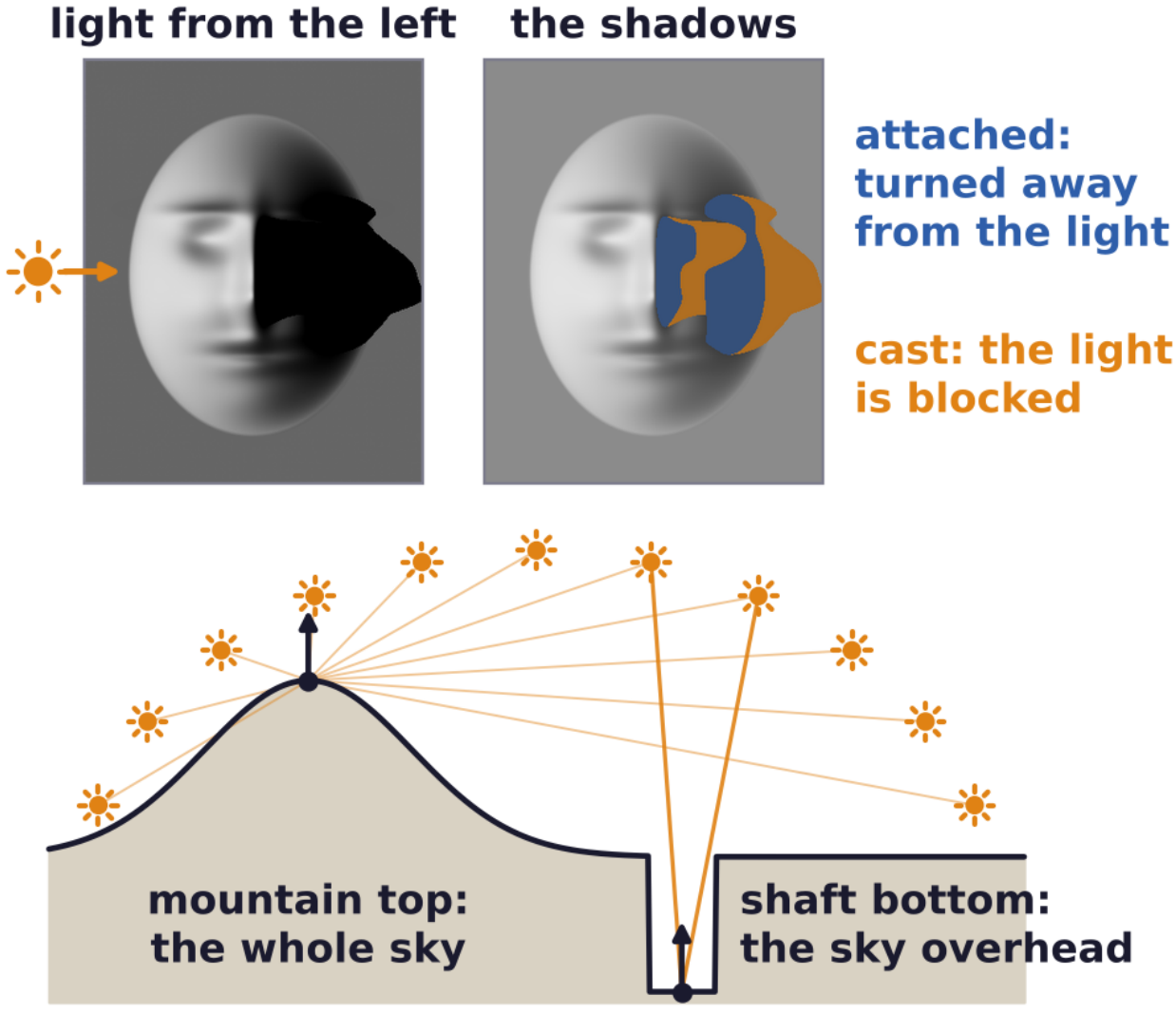



Figure 11: Top: the face lit from the left, 40° above the image plane, and its shadows (blue: attached; orange: cast). Bottom: a mountain top and a shaft bottom with the same normal, under a sky of lights; orange lines: the lights each point sees.

# With known lights, photometric stereo finds the normals

Photometric stereo takes several images of a Lambertian object from one viewpoint, each under a single directional (distant) light source, and recovers the normals and the albedo. Basri writes the model for all images and pixels at once as $M = LS$: the matrix $M$ $(f \times p)$ holds the $f$ images as rows of $p$ pixels; row $j$ of $L$ $(f \times 3)$ is the light $\vec{l}_j$ of image $j$; column $q$ of $S$ $(3 \times p)$ is the scaled normal $\vec{n}_q = \rho_q \hat{n}_q$ of pixel $q$. Entry $(j, q)$ of $LS$ is $\vec{l}_j^{\,T} \vec{n}_q$, Lambert's law.

If $L$ is known and $f \geq 3$ lights are not coplanar, each pixel gives $f$ linear equations in the three unknowns of $\vec{n}_q$, solved by least squares (Woodham, 1980):

$$\vec{n}_q = (L^T L)^{-1} L^T \, m_q,$$

where $m_q$ is column $q$ of $M$, the intensities of pixel $q$ in the $f$ images. Then $\rho_q = |\vec{n}_q|$ and $\hat{n}_q = \vec{n}_q / \rho_q$. The algorithm extends to more complex reflectance models, if they are known, through a lookup table from intensities to normals.

# With known lights, photometric stereo finds the normals

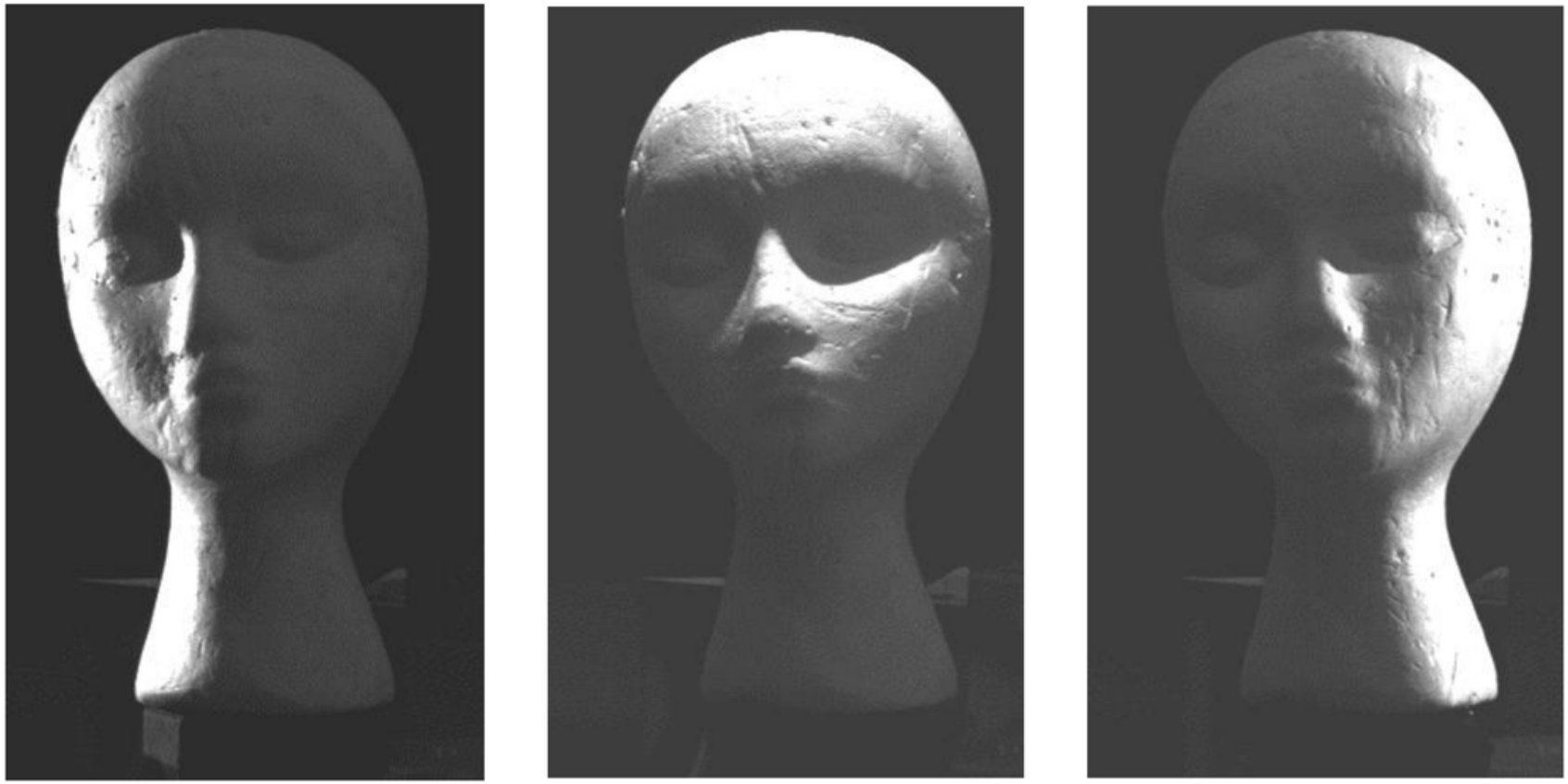


Figure 12: A white head lit from three directions, three of the $f$ images of photometric stereo; with their three known lights, each pixel's three intensities determine its scaled normal. B1 p. 19.

# With unknown lights, shape and lighting are fitted together

G estimates unknown lights together with the albedo and the shape (Yuille et al., 1999). It writes $\vec{b}(\vec{x}) = a(\vec{x})\vec{n}(\vec{x})$, so the albedo is $a(\vec{x}) = |\vec{b}(\vec{x})|$ and the normal is $\vec{n}(\vec{x}) = \vec{b}(\vec{x})/a(\vec{x})$, and fits $\vec{b}$ and the lights $\vec{s}^{\,\mu}$ to $N$ images $I^\mu$ by the Gibbs distribution

$$P(\vec{b}\,|\,I) = \frac{1}{Z}\exp\{-E[\vec{b}]\}, \qquad E[\vec{b}] = \sum_{\mu=1}^{N}\sum_{\vec{x}}\left(I^\mu(\vec{x}) - \vec{b}(\vec{x})\cdot\vec{s}^{\,\mu}\right)^2,$$

where $Z$ normalizes $P$ and the energy $E$ is the squared error (in Lambert's law $E$ was the light's strength); the most probable $\vec{b}$ minimizes $E$. In matrix form, $J$ (pixels $p$ by images $\mu$) holds the images as columns, $B$ (pixels by 3) the $\vec{b}$ as rows and $S$ ($3 \times N$) the lights, so $J = M^T$, $B = S^T$, $S = L^T$ in the letters of photometric stereo:

$$E[B,S] = \sum_{p,\mu}\left(J_{p\mu} - \sum_{i=1}^{3} B_{pi}S_{i\mu}\right)^2.$$

# With unknown lights, shape and lighting are fitted together

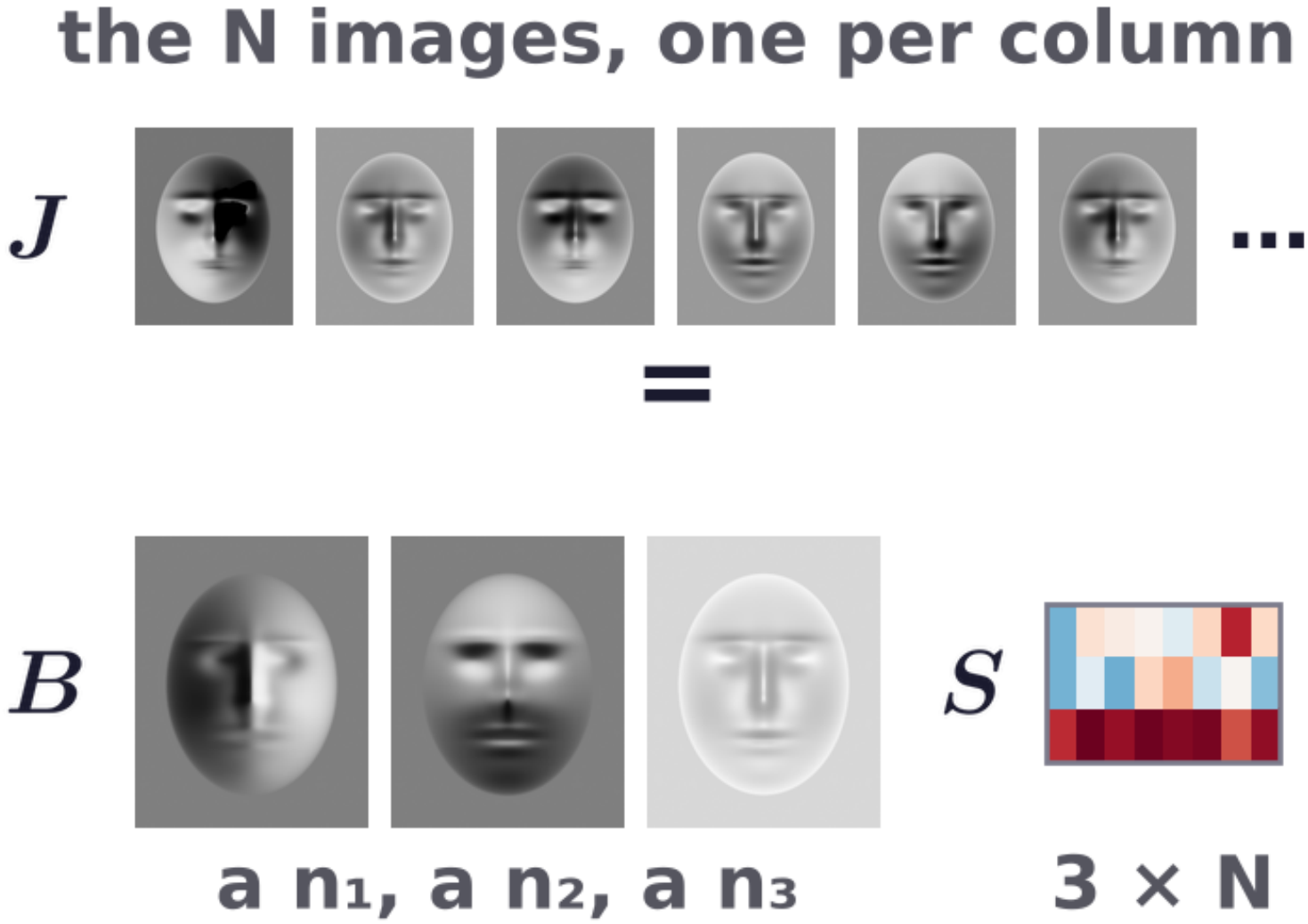



Figure 13: $J = BS$ for the synthetic face: six of the $N = 40$ images as columns of $J$ (drawn with shadows, so for them $J \approx BS$; next frame); the three columns of $B$ shown as images ($a\boldsymbol{n_1}$, $a\boldsymbol{n_2}$, $a\boldsymbol{n_3}$); the first eight columns of $S$, the $3 \times N$ lights (red positive, blue negative).

# The singular value decomposition (SVD) gives the rank-3 fit

Minimizing $E[B, S]$ asks for the matrix $BS$ of rank at most three closest to $J$ in the sum of squares. The singular value decomposition (SVD) writes $J = UDV^T$, with $U^TU = I$, $V^TV = I$ ($I$ the identity) and $D$ diagonal with $D_{11} \geq D_{22} \geq \cdots \geq 0$. Since $JJ^T = UD^2U^T$ and $J^TJ = VD^2V^T$, the columns of $U$ and $V$ are the eigenvectors of $JJ^T$ and $J^TJ$, with eigenvalues $D_{ii}^2$. Let $e_k(p)$ and $f_k(\mu)$, $k = 1, 2, 3$, be the first three columns of $U$ and $V$. The best rank-3 approximation keeps the three largest singular values (Eckart–Young), so the minimum is reached at

$$B_{pi} = \sum_{k=1}^{3} P_{ik} e_k(p), \quad S_{i\mu} = \sum_{k=1}^{3} Q_{ik} f_k(\mu), \quad P^TQ = \operatorname{diag}(D_{11}, D_{22}, D_{33}),$$

because then $(BS)_{p\mu} = \sum_k D_{kk} e_k(p) f_k(\mu)$. The leftover energy is $\sum_{i \geq 4} D_{ii}^2$. It tests the model: it is zero exactly when $J = BS$, which happens without shadows, because the images of a Lambertian object then lie in a three-dimensional space (Shashua, 1997). Hayakawa (1994) factorizes $M = LS$ in the same way, by the SVD of $M$.

# The singular value decomposition (SVD) gives the rank-3 fit

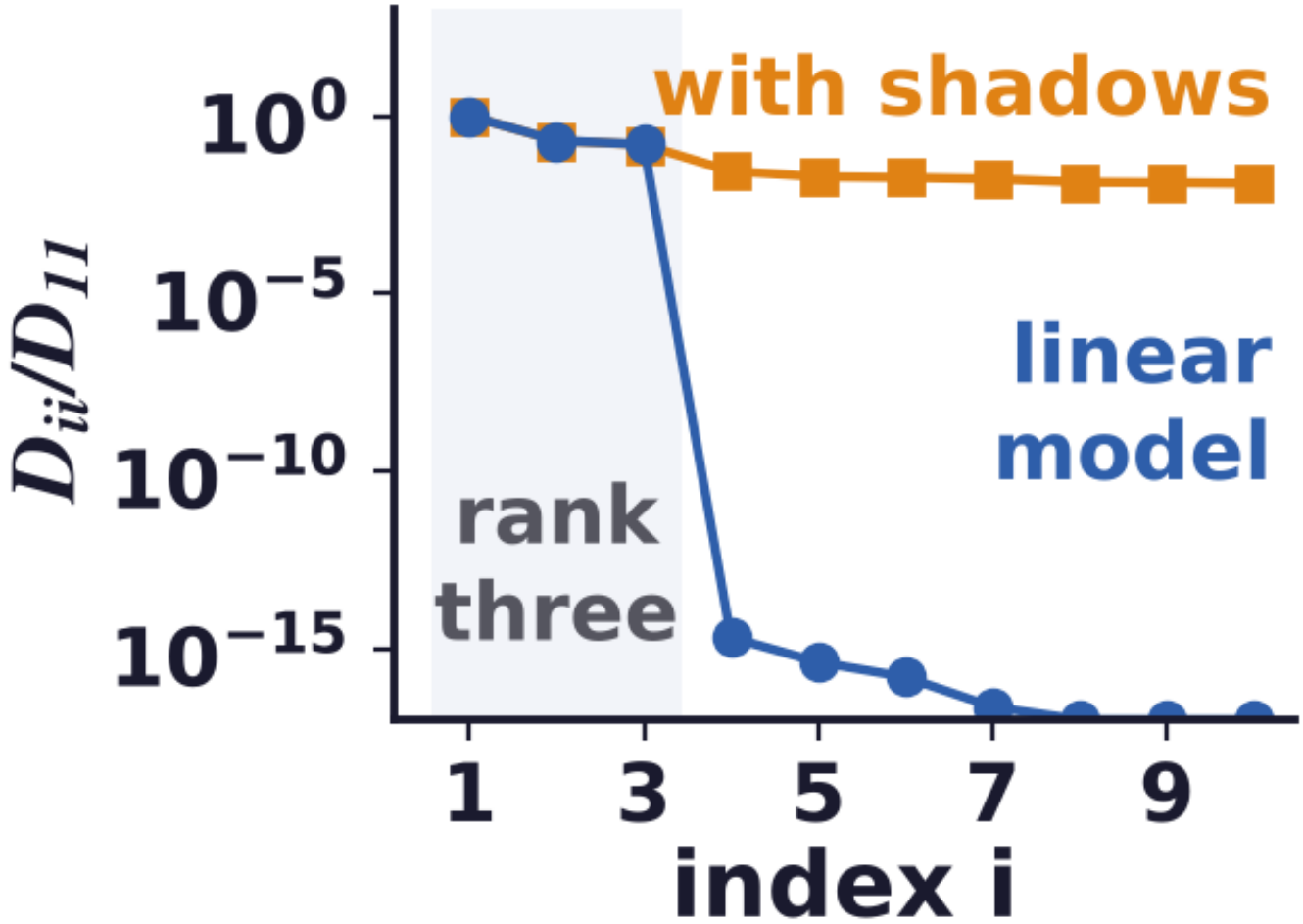



Figure 14: Singular values of $J$ for the synthetic face under $N = 40$ lights, relative to the largest, on a log scale. Blue: the linear model, rank three (the rest is rounding error, $10^{-15}$ to $10^{-16}$). Orange: with attached and cast shadows.

# Shape and lights are found only up to an invertible matrix

The solution fixes only the product $P^T Q$, and for any invertible $3 \times 3$ matrix $A$ the change $P \mapsto A^T P$, $Q \mapsto A^{-1} Q$ keeps it: $P^T A A^{-1} Q = P^T Q$. In terms of the surface and the lights,

$$\vec{b} \mapsto A^T \vec{b}, \qquad \vec{s}_\mu \mapsto A^{-1} \vec{s}_\mu$$

keeps every product, since $(A^T \vec{b}) \cdot (A^{-1} \vec{s}) = \vec{b}^T A A^{-1} \vec{s} = \vec{b} \cdot \vec{s}$. The ambiguity has nothing to do with the SVD: it is inherent in the Lambertian model, and since $\max\{\vec{b} \cdot \vec{s}_\mu, 0\}$ is unchanged too, it survives attached shadows. Cast shadows depend on the surface itself, and only a subgroup of these transformations keeps them (the generalized bas-relief).

The transformed $\vec{b}$ has other normals, $A^T \vec{b}/|A^T \vec{b}|$, and another albedo, $|A^T \vec{b}|$; with the lights $A^{-1} \vec{s}$ it gives the same images. This is similar to the ambiguity of structure from rigid motion, also solved by the SVD (Lecture 16): there, too, the factorization fixes the shape only up to a $3 \times 3$ matrix.

# Shape and lights are found only up to an invertible matrix

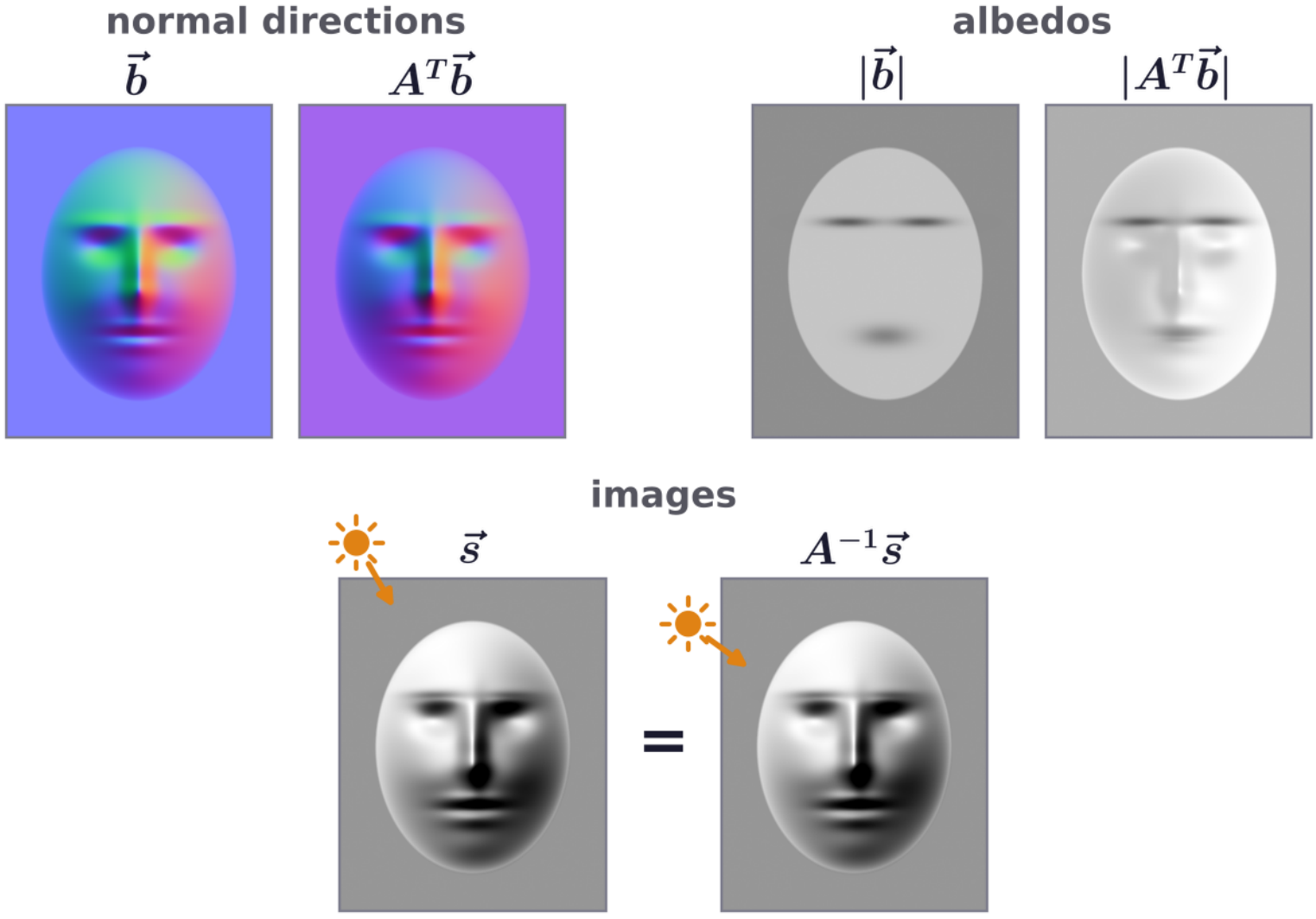



Figure 15: Top: the normals of $\vec{b}$ and of $A^T\vec{b}$ (colors as on the key sphere of the model frame), and their albedos $|\vec{b}|$ and $|A^T\vec{b}|$. Bottom: the images under $\vec{s}$ and under $A^{-1}\vec{s}$ are identical (attached shadows included).

# Integrability: a surface returns to its height around a loop

One constraint reduces the ambiguity: after a closed loop on a surface you are back at the height you started from. For a surface $(x, y, z(x, y))$ the vectors $(1, 0, z_x)$ and $(0, 1, z_y)$ are tangent to it, and the normal, orthogonal to both, is

$$\vec{n} = (n_1, n_2, n_3) = \frac{(-z_x, -z_y, 1)}{\sqrt{z_x^2 + z_y^2 + 1}}, \qquad \text{so} \quad \frac{n_1}{n_3} = -z_x, \quad \frac{n_2}{n_3} = -z_y.$$

Because $z_{xy} = z_{yx}$ (the order of differentiation does not matter), the normals of a real surface satisfy the integrability condition

$$\frac{\partial}{\partial x}\frac{n_2}{n_3} = \frac{\partial}{\partial y}\frac{n_1}{n_3},$$

the differential form of the closed-loop rule.

# Integrability: a surface returns to its height around a loop

Given estimated normals, the depth follows by least squares (B1 pp. 24–25), with the slopes $p = -n_1/n_3 = z_x$, $q = -n_2/n_3 = z_y$ (these $p, q$ are slopes, not pixel indices):

$$\min_z \sum_{x,y} \big(z(x{+}1, y) - z(x, y) - p\big)^2 + \big(z(x, y{+}1) - z(x, y) - q\big)^2.$$

The residual $\partial_x(n_2/n_3) - \partial_y(n_1/n_3)$ is zero for the normals of a surface, such as the face or a flattened and tilted face (a bas-relief), and large for $A^T\vec{b}$ after a general $A$.




Figure 16: The integrability residual for the true normals, after the general $A$ and after a bas-relief transformation, on one color scale (red positive, blue negative, white zero).

# Integrability leaves the generalized bas-relief ambiguity

Integrability is consistent only with a subgroup of the invertible linear transformations (Belhumeur et al., 1999):

$$b_1 \mapsto \lambda b_1 + \alpha b_3, \qquad b_2 \mapsto \lambda b_2 + \beta b_3, \qquad b_3 \mapsto \tau b_3,$$

with constants $\lambda, \alpha, \beta, \tau$ ($\lambda, \tau \neq 0$). Dividing, the new slopes are

$$-z'_x = \frac{b'_1}{b'_3} = \frac{\lambda b_1 + \alpha b_3}{\tau b_3} = \frac{\lambda}{\tau}(-z_x) + \frac{\alpha}{\tau}, \qquad -z'_y = \frac{\lambda}{\tau}(-z_y) + \frac{\beta}{\tau},$$

using $b_1/b_3 = -z_x$. Integrating (a constant does not change the image),

$$z'(x,y) = \lambda' z(x,y) + \mu x + \nu y, \qquad \lambda' = \frac{\lambda}{\tau}, \quad \mu = -\frac{\alpha}{\tau}, \quad \nu = -\frac{\beta}{\tau}$$

The term $\lambda' z$ is a bas-relief transformation: a flattening ($0 < \lambda' < 1$), a deepening ($\lambda' > 1$) or, for $\lambda' < 0$, an inversion of the surface; $\mu x + \nu y$ adds a plane, which tilts it ($\mu, \nu$: constants, not the light index). The result is a surface, hence integrable; a general $A$ mixes $b_1$ and $b_2$ and breaks integrability.

# Integrability leaves the generalized bas-relief ambiguity

The whole effect is the generalized bas-relief (GBR) ambiguity. Write $M$ for the matrix of the transformation of $\vec{b}$ (rows $(\lambda, 0, \alpha)$, $(0, \lambda, \beta)$, $(0, 0, \tau)$). The images stay the same if each light becomes $M^{-T}\vec{s}$, because $(M\vec{b}) \cdot (M^{-T}\vec{s}) = \vec{b} \cdot \vec{s}$. A short computation gives $M^{-T} = G/\lambda$ with

$$G = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \mu & \nu & \lambda' \end{pmatrix},$$

the matrix that maps each surface point $(x, y, z)$ to $(x, y, z')$. So the lights move to $G\vec{s}$, up to a factor $1/\lambda$ that also rescales the albedo.

Because the same $G$ moves the points and the lights, a ray from a point toward the light that grazes the surface still grazes it: cast shadows are preserved as well (Belhumeur et al., 1999). The GBR therefore cannot be resolved by shadows, by any number of images taken from one viewpoint, or by integrability. The construction also extends to perspective projection.

# Integrability leaves the generalized bas-relief ambiguity

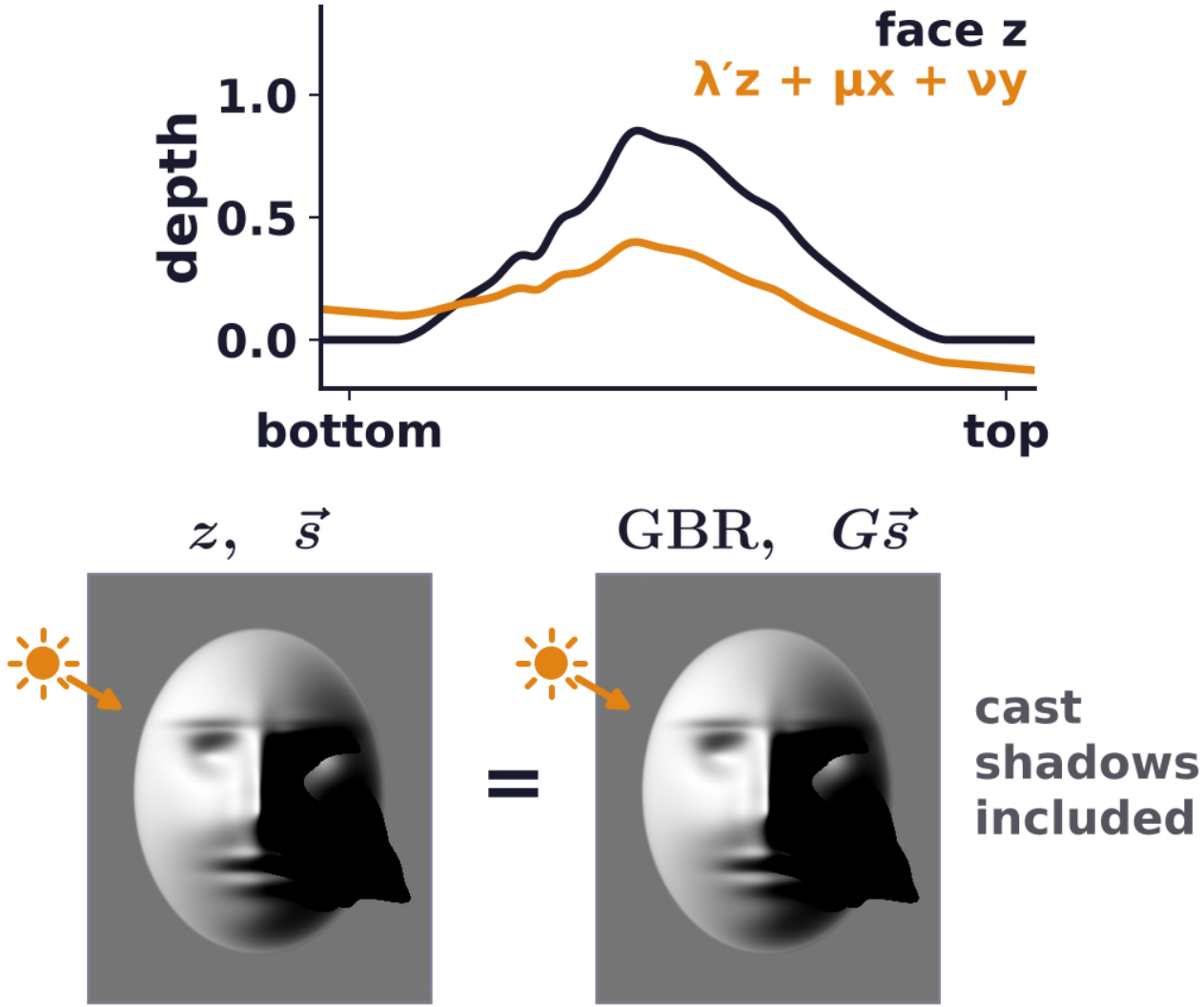



Figure 17: Top: profiles along the vertical midline of the face (black) and of its GBR transform with $\lambda' = 0.45$, $\mu = 0.15$, $\nu = -0.10$ (orange). Bottom: the face under $\vec{s}$ and the flattened, tilted face under $G\vec{s}$ give the same image, cast shadows included.

# Sculptors and human vision show the bas-relief ambiguity

Sculptors have long used the bas-relief ambiguity to save material: a relief, with small changes in depth, looks like a full figure when seen from the front. Make-up can alter the albedo of a face and make its shape appear different, for instance by enhancing the cheek bones. Studies of how people estimate shape (by Bülthoff, Koenderink and others) find biases in perceived shape; Koenderink et al. (2001) found biases like the bas-relief ambiguity.

Yuille et al. (2003) extend the GBR to changes of viewpoint, the KGBR: an affine transformation of the shape (a linear map plus a translation) together with a matching change of the albedo. For two objects related by a KGBR one can always find corresponding viewpoints and lightings for which they look identical, the images agreeing up to an affine warp of the image (for orthographic projection). The GBR is the special case in which both objects are seen from the same viewpoint, each under its own lighting.

# Sculptors and human vision show the bas-relief ambiguity

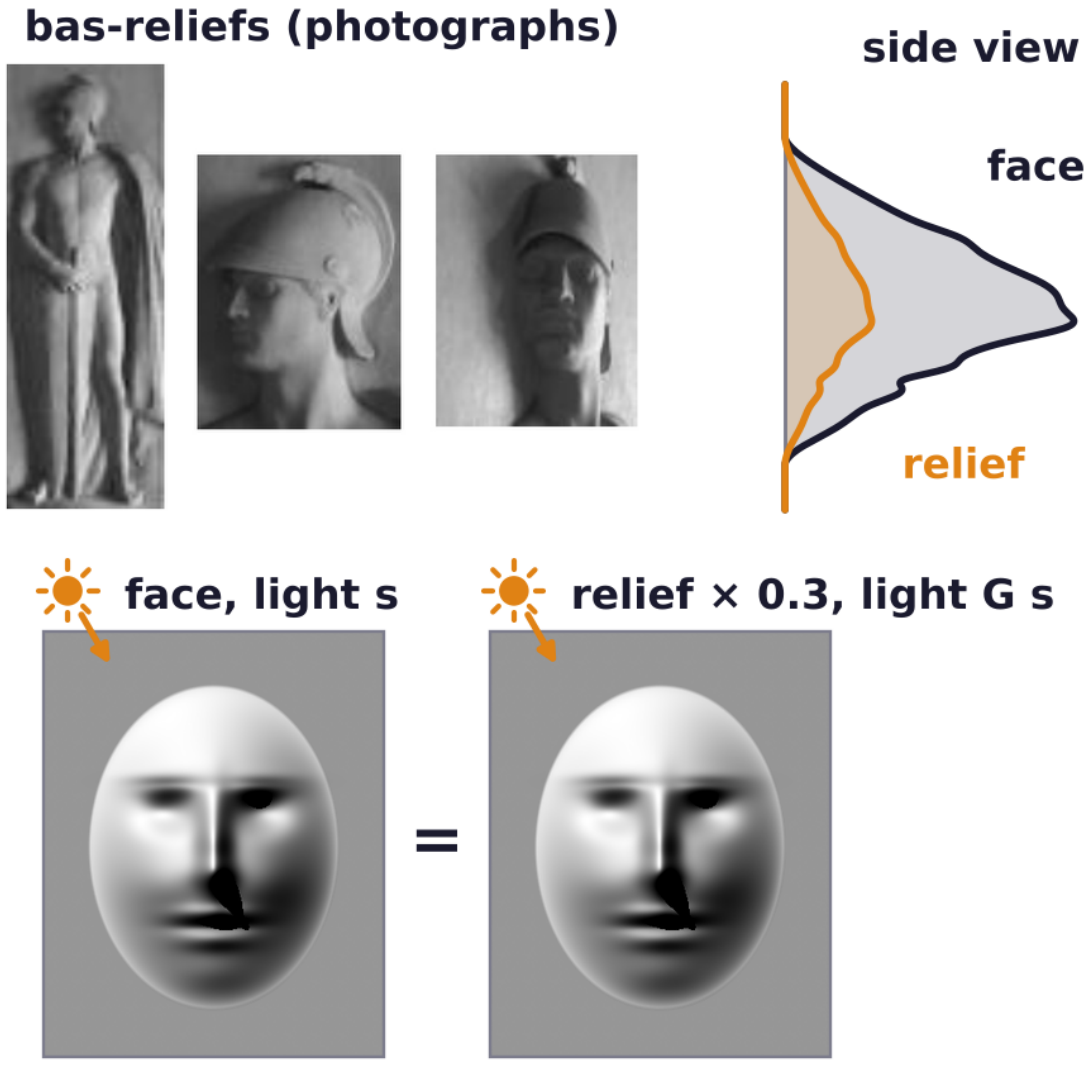



Figure 18: Top: bas-reliefs (B1 p. 23), and the side view of the face and of its bas-relief with depth $\times 0.3$. Bottom: the face under $\vec{s}$ and the relief under $G\vec{s}$ (same azimuth, lower) look the same.

# Shape from shading estimates shape from a single image

Humans can estimate the shape of an object from its shading pattern in a single image, but for a computer this is a severely ill-posed problem: a problem whose data do not determine the solution. Current methods make strong assumptions, such as a constant albedo and a known light source.

With a known light $\vec{l}$ and a uniform albedo, the intensity is $I = \vec{l}^{\,T}\vec{n} \propto \cos\theta$, the cosine of the angle between the normal and the light. Every intensity therefore determines a cone of possible normals around the light direction: on the sphere of unit normals, a circle. The normal has two unknowns (its direction), and the intensity gives one equation. Writing the normal through the depth, $\vec{n} \propto (-z_x, -z_y, 1)$, leaves one unknown per pixel, the depth $z$, since uniform albedo is assumed; the neighbors couple the pixels through the derivatives. Classic methods (Horn, 1975) assume known light directions and a locally smooth surface.

# Shape from shading estimates shape from a single image

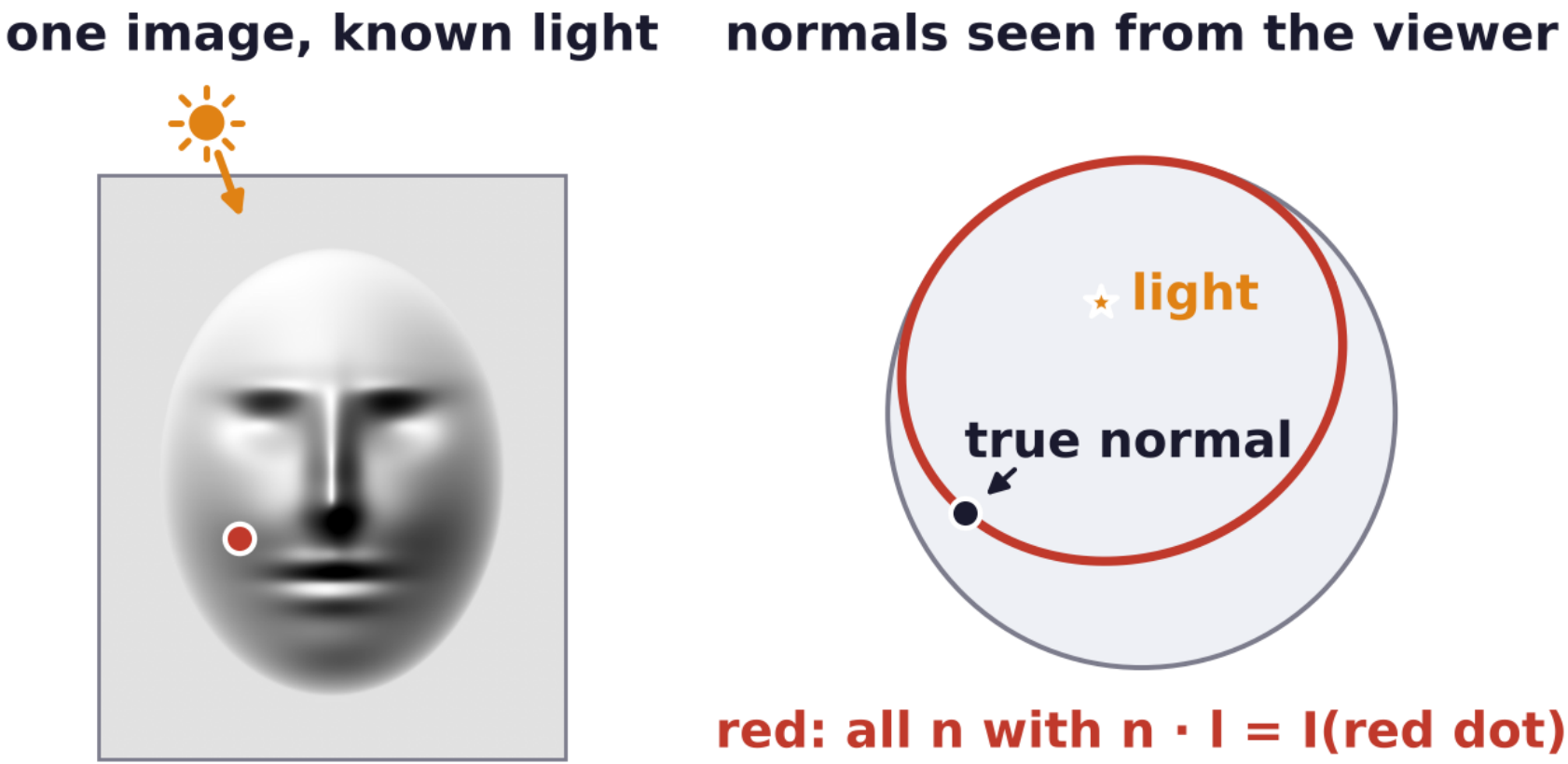



Figure 19: One image of the face with uniform albedo under a known light; the red dot is one pixel. Beside it, the hemisphere of normals seen from the viewer, with the light (star); the red circle holds every normal that gives the red dot's intensity, the true normal among them.

# Shape from shading is a first-order nonlinear PDE

Write the normal before normalizing, $\vec{n} = (-z_x, -z_y, 1)$, and the image intensity as $E$ (Basri's letter; $E$ was the light's strength in Lambert's law and G's energy). With uniform albedo 1,

$$E = \frac{\vec{l}^T \vec{n}}{\|\vec{n}\|} \;\Rightarrow\; E^2\, \vec{n}^T \vec{n} = \vec{n}^T \vec{l}\,\vec{l}^T \vec{n} \;\Rightarrow\; \vec{n}^T \left(\vec{l}\,\vec{l}^T - E^2\, \mathrm{I}\right) \vec{n} = 0,$$

where $\mathrm{I}$ is the $3 \times 3$ identity matrix. This is a first-order, nonlinear partial differential equation (PDE) for $z$ (Horn, 1975): first-order because only $z_x$ and $z_y$ appear, nonlinear because they appear squared.

For a light along the viewing direction, $\vec{l} = (0, 0, 1)$, it becomes

$$E^2(z_x^2 + z_y^2 + 1) = 1 \quad \Leftrightarrow \quad z_x^2 + z_y^2 = \frac{1}{E^2} - 1,$$

an eikonal equation, $\|\nabla z\| = f$ with $f = \sqrt{1/E^2 - 1}$, the equation of a front that moves with speed $1/f$. The image thus gives the slope $|\nabla z|$ at every pixel; the direction of steepest ascent is the remaining unknown.

# Shape from shading is a first-order nonlinear PDE

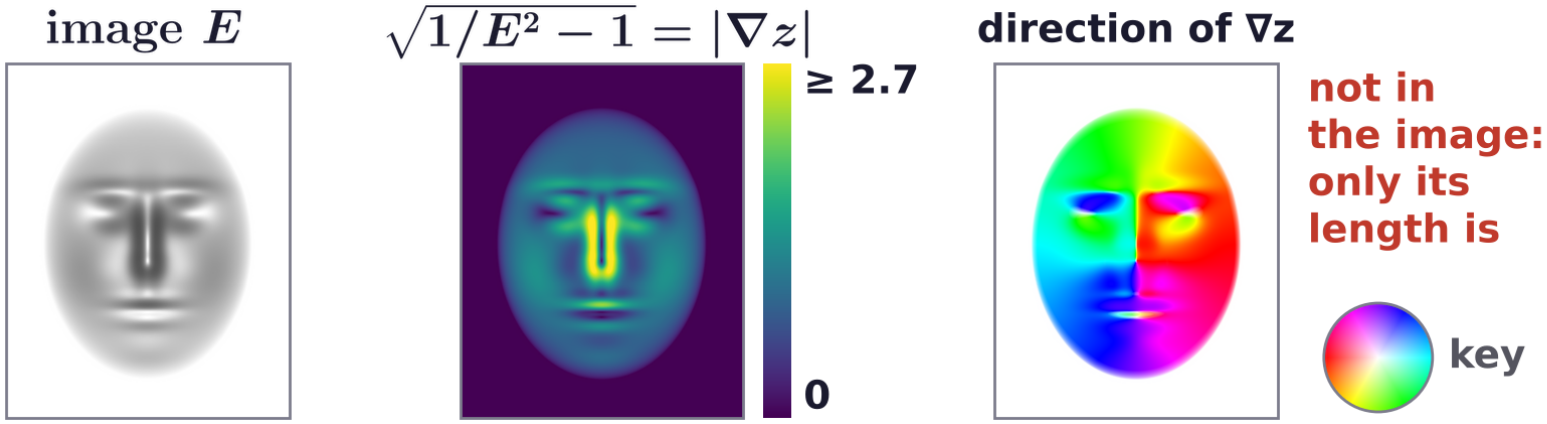



Figure 20: For $\vec{l} = (0, 0, 1)$: the image $E$; the slope $\sqrt{1/E^2 - 1}$ computed from it, which equals $|\nabla z|$ (color scale saturated at 2.7, the 99.5th percentile; the steepest slope is 2.9); the direction of $\nabla z$ (hue, with its key), which is not in the image.

Bright pixels face the light and have small slopes; the slope is largest along the sides of the nose and grows toward the rim of the face.

# Fast marching solves the eikonal equation by shortest paths

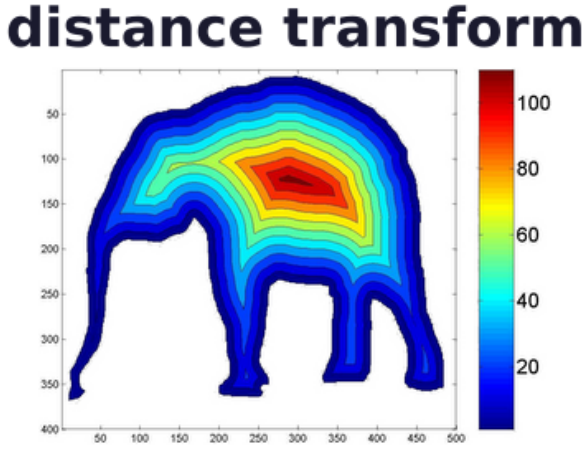
The distance transform gives each point of a region its distance to the boundary. It solves the eikonal equation $\|\nabla z\|^2 = 1$ with $z = 0$ on the boundary. On a graph the analogous quantity is a shortest-path distance: a node's distance $T$ is the minimum over its neighbors of their $T$ plus the weight $w$ of the edge, $T = \min\{T_1 + w_1, T_2 + w_2\}$, and Dijkstra's algorithm fixes the nodes in increasing order of $T$.

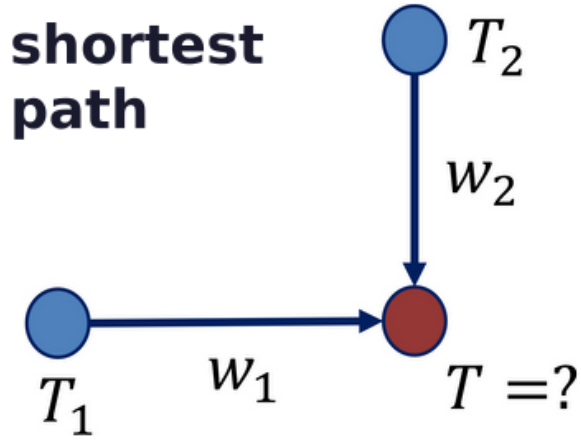
Fast marching (Tsitsiklis, 1995; Sethian, 1996) does the same on a grid for $\|\nabla T\| = f$, with $f$ the local cost. At a grid point whose fixed neighbors are $T_1$ (horizontal) and $T_2$ (vertical), on a unit grid, the discretized equation and its larger root are

$$(T - T_1)^2 + (T - T_2)^2 = f^2, \qquad T = \tfrac{1}{2}\Big(T_1 + T_2 + \sqrt{2f^2 - (T_1 - T_2)^2}\Big).$$

This is valid when $T \geq \max(T_1, T_2)$, which holds exactly when $|T_1 - T_2| \leq f$; otherwise the front comes from one neighbor only, and $T = f + \min\{T_1, T_2\}$. Nodes are fixed in increasing order of $T$, as in a shortest-path algorithm.

# Fast marching solves the eikonal equation by shortest paths




$$\|\nabla z\|^2 = z_x^2 + z_y^2 = 1$$

$$T = \min\{T_1 + w_1, T_2 + w_2\}$$

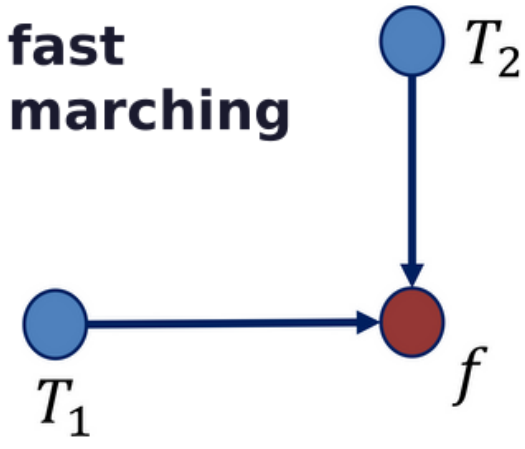



Figure 21: The distance transform of an elephant (color: the distance to the boundary); the shortest-path update at a node (red) from two neighbors (blue) and edge weights $w_1, w_2$; the fast-marching update from two grid neighbors and the local cost $f$. After B1 pp. 30–34, whose update lacks the factor $\frac{1}{2}$ and is used whenever the root is real; it holds only for $|T_1 - T_2| \leq f$.

# Fast marching solves Lambertian shape from shading

Kimmel and Sethian (2001) solve Lambertian shape from shading by fast marching, with the local cost $f = \sqrt{1/E^2 - 1}$ computed from the image. Boundary conditions are depth values at the local maxima of intensity (points that face the light, where $f = 0$) and possibly at the shape boundaries; a light $\vec{l} \neq (0, 0, 1)$ is handled by a change of variables.

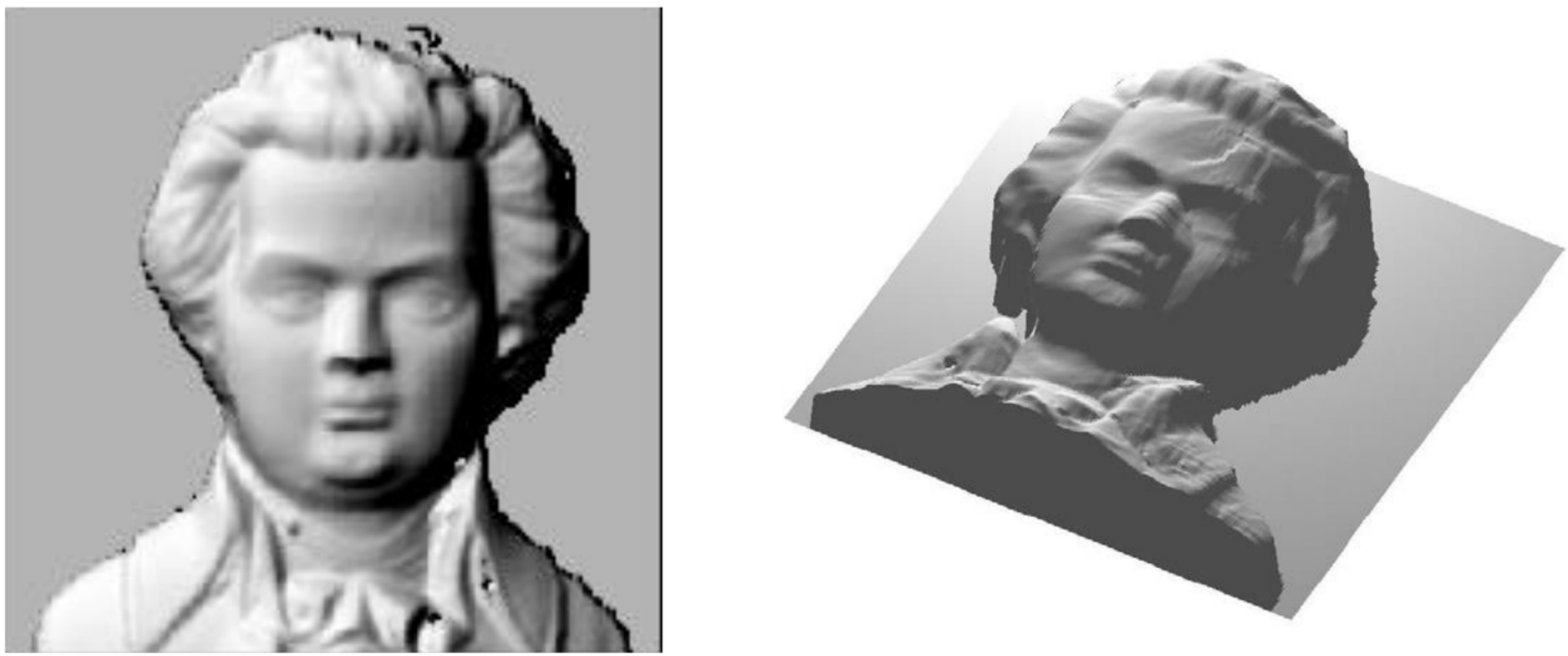


Figure 22: An image of a bust of Mozart, a standard synthetic test image (Zhang et al., 1999), and the surface recovered from it, rendered from the side. B1 p. 36.

# Assumptions or priors resolve the ambiguities of one image

Shape from shading methods typically assume that the albedo is constant and the light source is known; this is often sufficient for a unique solution (Oliensis, 1991). Other methods make strong assumptions about the object. Atick et al. (1996) use a prior on the shape of faces: principal components of three-dimensional (3D) face depth maps, so a face is the mean depth map plus a few weighted components (eigenheads). Others assume something about the light directions or the albedo, for example that the albedo is piecewise smooth.

Statistical shape from shading learns the relation between intensity and depth from range data, images taken together with a laser range finder (Potetz and Lee, 2003). Their results suggest that Lambertian models rarely apply in natural images (partly because of cast shadows, partly because of atmospheric effects): in general, objects nearer to the viewer tend to look brighter.

# Assumptions or priors resolve the ambiguities of one image

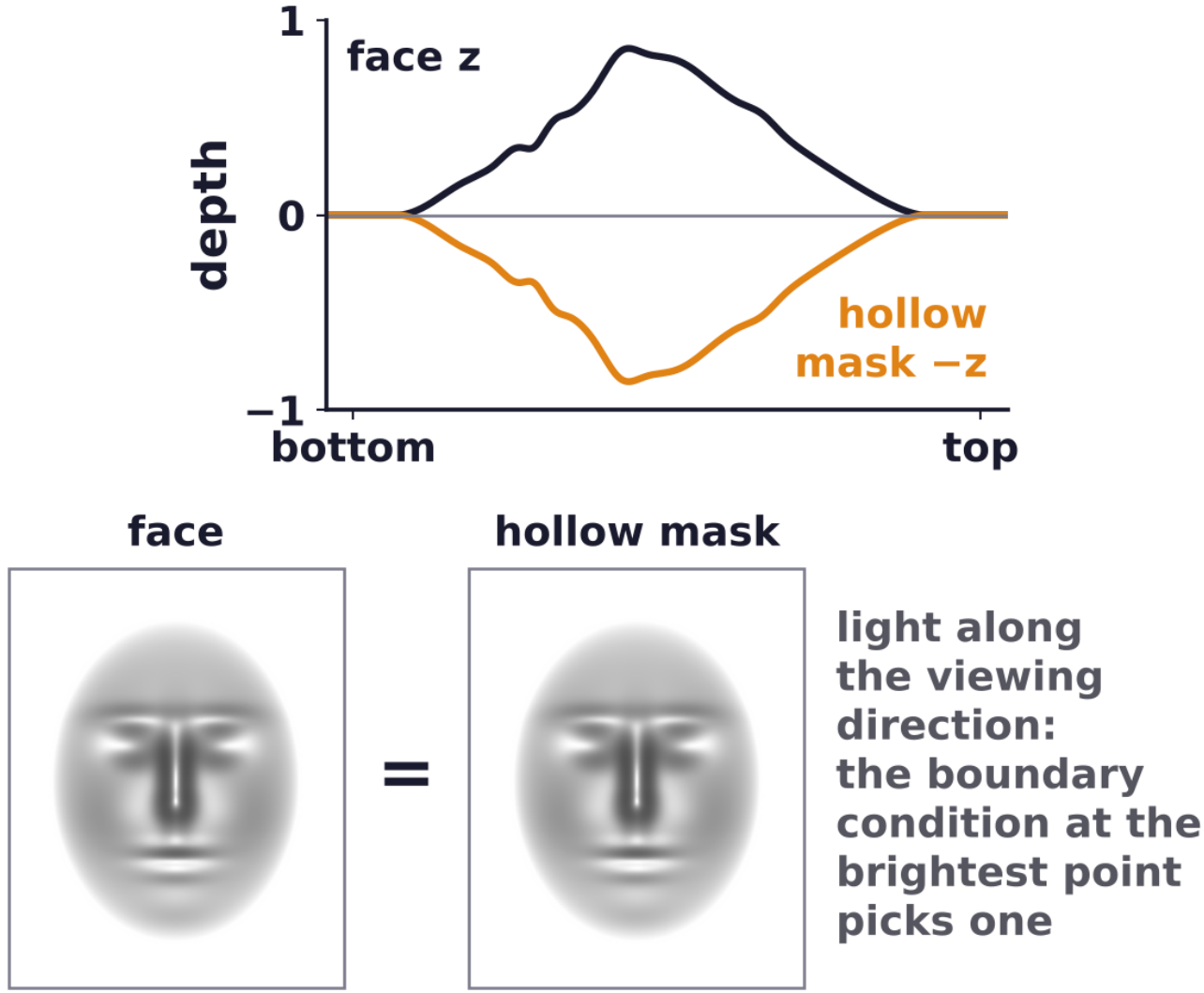



Figure 23: Top: the profile of the face $z$ and of the hollow mask $-z$. Bottom: under a light along the viewing direction the two give the same image; only the boundary condition at the brightest point (bump or dip) picks one.

# The images under all lightings form a convex cone

What is the set of images of an object under all lightings, with any number of light sources? Images are additive and non-negative, so any non-negative combination of images is again an image: the set is a convex cone in $\mathbb{R}^p$, the space of images of $p$ pixels (Belhumeur and Kriegman, 1998). The characterization is generic: it also holds with specularities, shadows and inter-reflections. Representing the cone is complicated (it has infinitely many degrees of freedom), but for Lambertian objects it is thin: a few directions hold most of it.

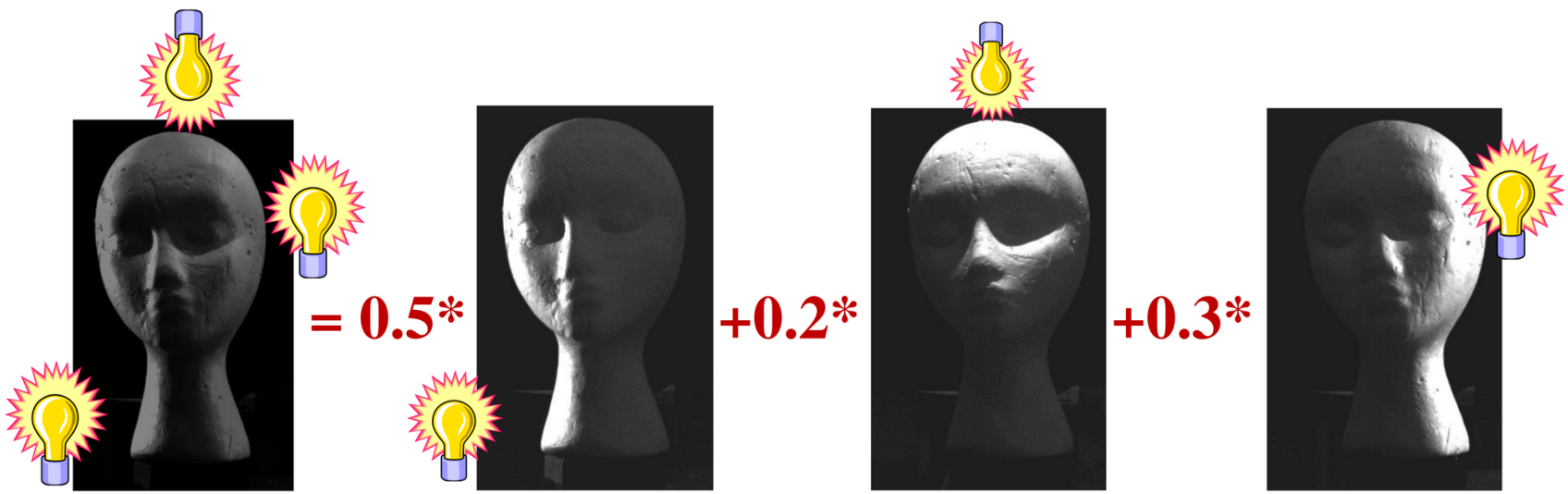



Figure 24: An image under three lights equals 0.5, 0.2 and 0.3 times the images under each light. B2 p. 2.

# A few principal components capture the images of an object

The thinness can be measured by principal component analysis (PCA). Photograph an object under $N$ lightings, $I^\mu$, $\mu = 1, \ldots, N$, and form the correlation of pixels

$$K(\vec{x}, \vec{x}') = \frac{1}{N} \sum_{\mu=1}^{N} I^\mu(\vec{x})\, I^\mu(\vec{x}').$$

Its eigenvectors, $\sum_{\vec{x}'} K(\vec{x}, \vec{x}') e(\vec{x}') = \lambda\, e(\vec{x})$, are images, the eigenimages, and each eigenvalue $\lambda$ is the energy (the mean squared projection of the images) along its eigenimage. The first $k$ eigenimages hold the fraction $\sum_{i \le k} \lambda_i / \sum_i \lambda_i$ of the energy, and every image is close to a combination of them.

The table of Yuille et al. (1999) gives this fraction for four objects: with five eigenimages it is 85–98 %. Theory says that nine suffice for convex objects, and that in a single view about five or six suffice (Ramamoorthi, 2002). A Lambertian object without shadows needs only three, as the synthetic face shows.

# A few principal components capture the images of an object

| | Ball | Face | Phone | Parrot |
|---|---|---|---|---|
| #1 | 48.2 | 53.7 | 67.9 | 42.8 |
| #3 | 94.4 | 90.2 | 88.2 | 76.3 |
| #5 | 97.9 | 93.5 | 94.1 | 84.7 |
| #7 | 99.1 | 95.3 | 96.3 | 88.5 |
| #9 | 99.5 | 96.3 | 97.2 | 90.7 |

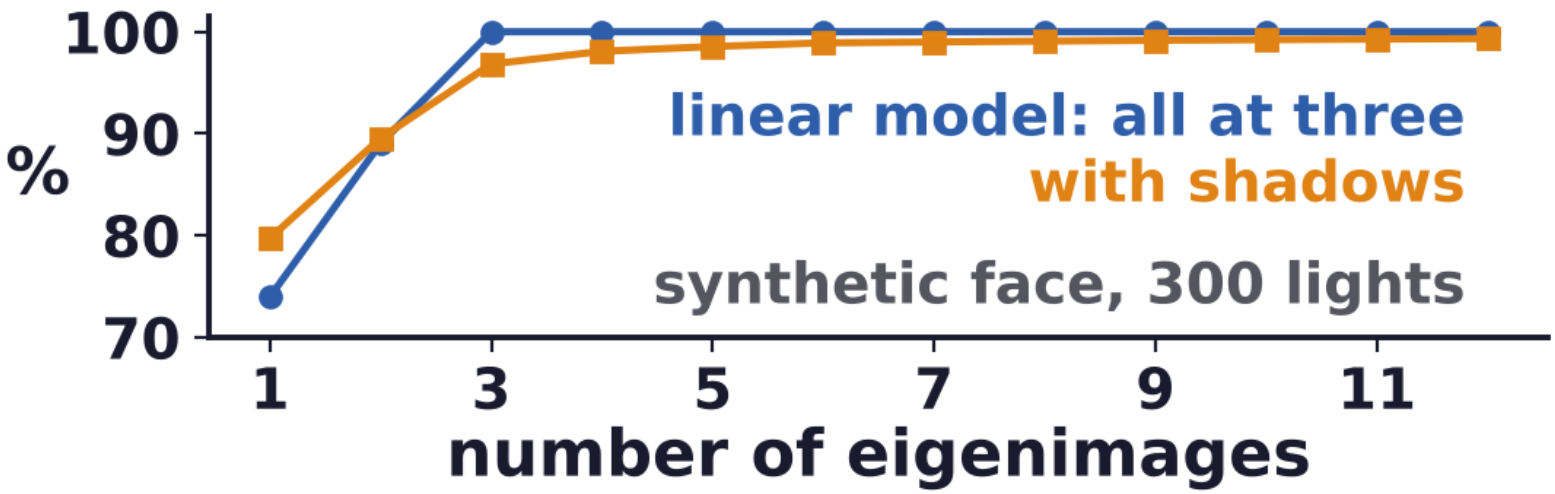



Figure 25: Top: percentage of the energy in the first $k$ eigenimages (row $\#k$) for a ball, a face, a phone and a parrot (B2 p. 4). Bottom: the synthetic face under 300 lights: blue the linear model, all at three; orange with shadows.

# Reflectance is the lighting convolved with a smooth kernel

Basri and Jacobs (2003) and Ramamoorthi and Hanrahan (2001) deal with attached shadows for convex objects, whose points each see all the lighting in front of them: the reflected light depends only on the normal $v \in S^2$ (the unit sphere). With $l(u)$ the light from $u$,

$$R(v) = \int_{S^2} k(u, v)\, l(u)\, du, \qquad k(u, v) = \max(\cos\theta_{uv}, 0),$$

with $\theta_{uv}$ the angle between $u$ and $v$. This Lambertian kernel depends only on that angle, so $R$ is a convolution on the sphere.

The kernel $k(u, v)$ is the light reflected at the normal $v$ per unit of light from $u$. Three point lights give a smooth reflectance: each adds its cosine, cut at zero when the light is behind the surface.

# Reflectance is the lighting convolved with a smooth kernel

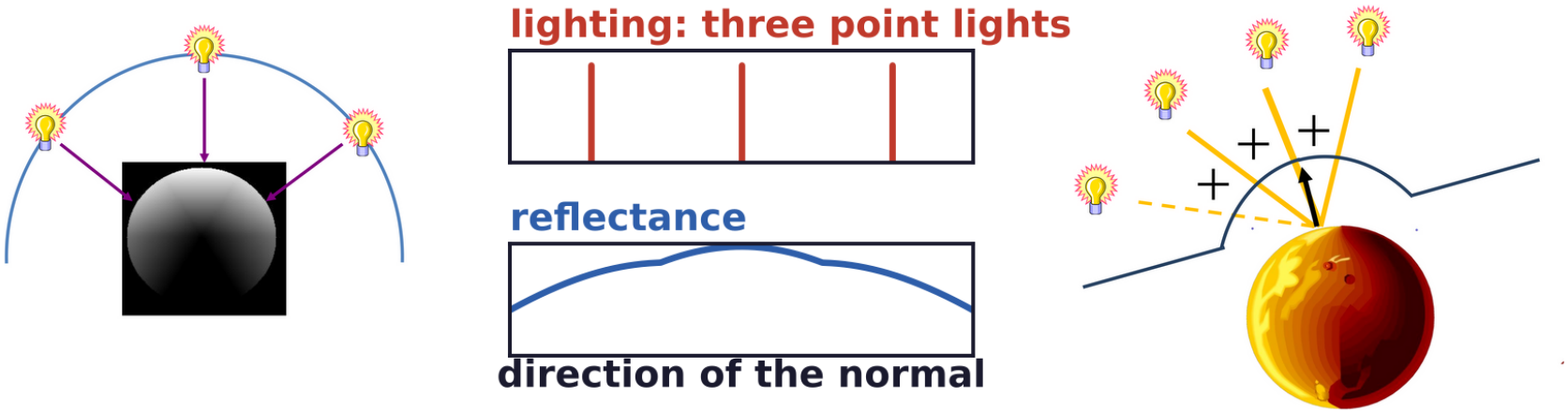



Figure 26: A sphere lit by three point lights (B2 p. 5); the lighting, three spikes on a circle of directions, and the smooth reflectance against the normal's direction; and the reflectance at one normal (arrow), which adds the cosine of each light in front of its tangent plane (+), while the light behind it (dashed) adds nothing (B2 p. 7).

# Spherical harmonics are an orthonormal basis on the sphere

Spherical harmonics are the analogue of sinusoids for functions on the sphere. With $\theta$ the polar angle and $\phi$ the azimuth,

$$Y_{nm}(\theta,\phi) = \sqrt{\frac{2n+1}{4\pi}\,\frac{(n-|m|)!}{(n+|m|)!}}\,P_{n|m|}(\cos\theta)\,e^{im\phi},$$

$$P_{nm}(z) = \frac{(1-z^2)^{m/2}}{2^n n!}\,\frac{d^{\,n+m}}{dz^{n+m}}(z^2-1)^n,$$

for $n = 0, 1, 2, \ldots$ and $m = -n, \ldots, n$; the $P_{nm}$ are the associated Legendre functions. The harmonics are orthonormal on the sphere, and the order $n$ has $2n+1$ of them. A rotation of the sphere mixes only harmonics of the same order; the shift $\phi \mapsto \phi + \phi_0$ about the polar axis multiplies $Y_{nm}$ by $e^{im\phi_0}$, a phase shift. In space coordinates $(X, Y, Z)$ on the unit sphere, real combinations of the harmonics of order $n$ are polynomials of degree $n$: $1$; $Z, X, Y$; $3Z^2-1$, $X^2-Y^2$, $XY$, $XZ$, $YZ$. The convolution theorem on the sphere (Funk–Hecke) makes them useful: convolving with a kernel that depends only on the angle multiplies each harmonic coefficient by a number that depends only on $n$.

# Spherical harmonics are an orthonormal basis on the sphere

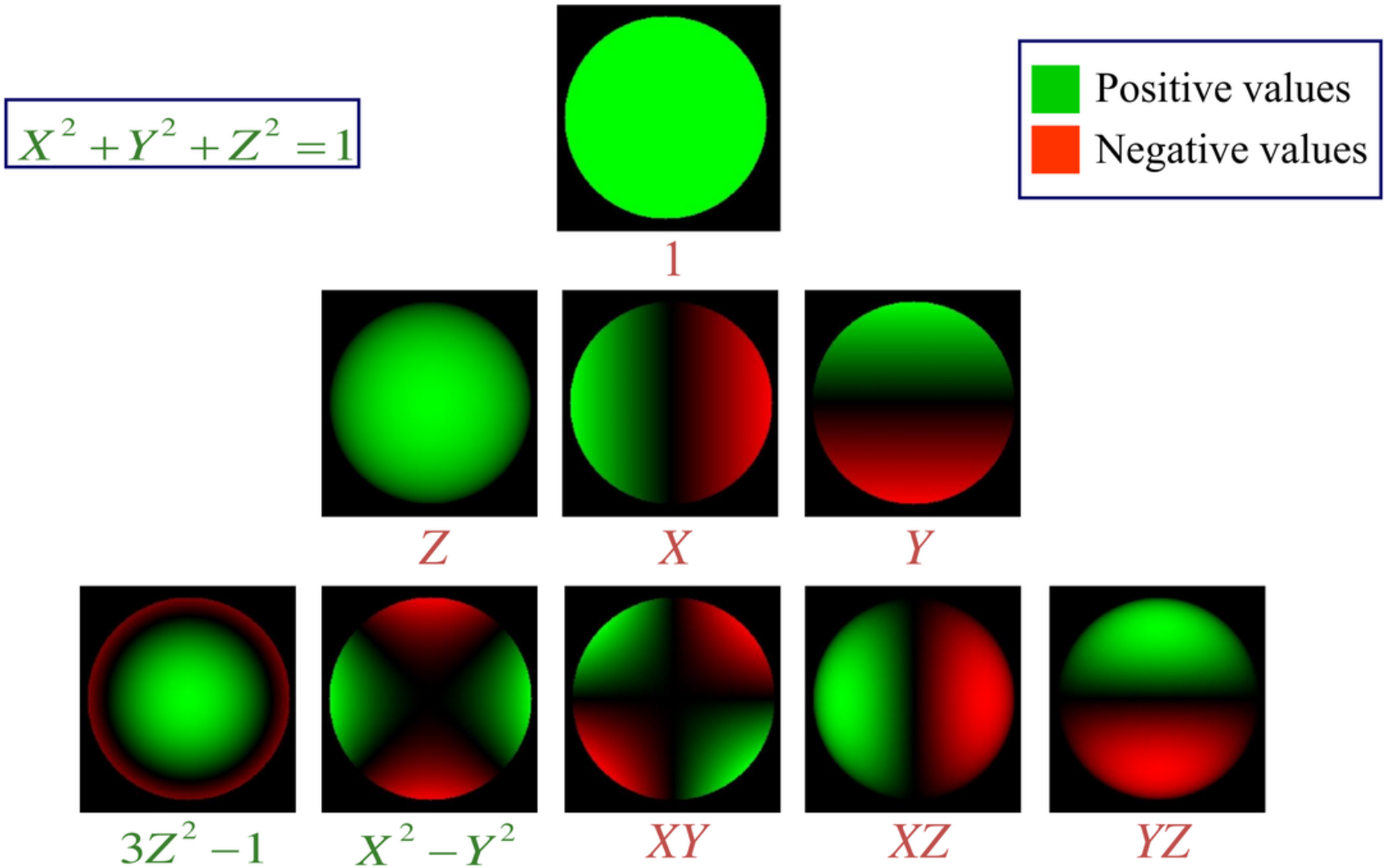



Figure 27: The nine harmonics of orders 0, 1, 2 on the sphere $X^2 + Y^2 + Z^2 = 1$ (green positive, red negative); in this picture $X$ increases to the left. B2 p. 9.

# Nine harmonics capture 99.2 % of the Lambertian kernel

Expand the lighting as $\ell = \sum_{n,m} l_{nm} Y_{nm}$ and the kernel as $k(\theta) = \max(\cos\theta, 0) = \sum_n k_n Y_{n0}(\theta)$. By the Funk–Hecke theorem the reflectance $r = k * \ell$ has the coefficients $\alpha_n l_{nm}$:

$$r(\theta,\phi) \approx \sum_{n=0}^{2} \sum_{m=-n}^{n} \alpha_n l_{nm} Y_{nm}(\theta,\phi), \qquad \alpha_n = \sqrt{4\pi/(2n+1)}\, k_n$$

(B2 and G write $k_n$ for $\alpha_n$). Computing $k_n = \int k\, Y_{n0}$ gives $k_0 = \sqrt{\pi}/2 = 0.886$, $k_1 = \sqrt{\pi/3} = 1.023$, $k_2 = \sqrt{5\pi}/8 = 0.495$, $k_3 = 0$, $k_4 = -0.111$, and $\alpha_0 = \pi$, $\alpha_1 = 2\pi/3$, $\alpha_2 = \pi/4$. The energy of the kernel is $\int_{S^2} k^2 = 2\pi/3 = \sum_n k_n^2$; orders 0–2 hold $(\pi/4 + \pi/3 + 5\pi/64)/(2\pi/3)$, that is 99.2 % of it, and orders 0, 1 hold 87.5 %. A point light's reflectance is the kernel itself, so nine basis images (orders $\leq 2$) give it with 99.2 % accuracy and four (ambient plus a point source) with 87.5 %; for any non-negative lighting the accuracy is at least 97.96 % and 75 % (Basri and Jacobs, 2003).

# Nine harmonics capture 99.2 % of the Lambertian kernel

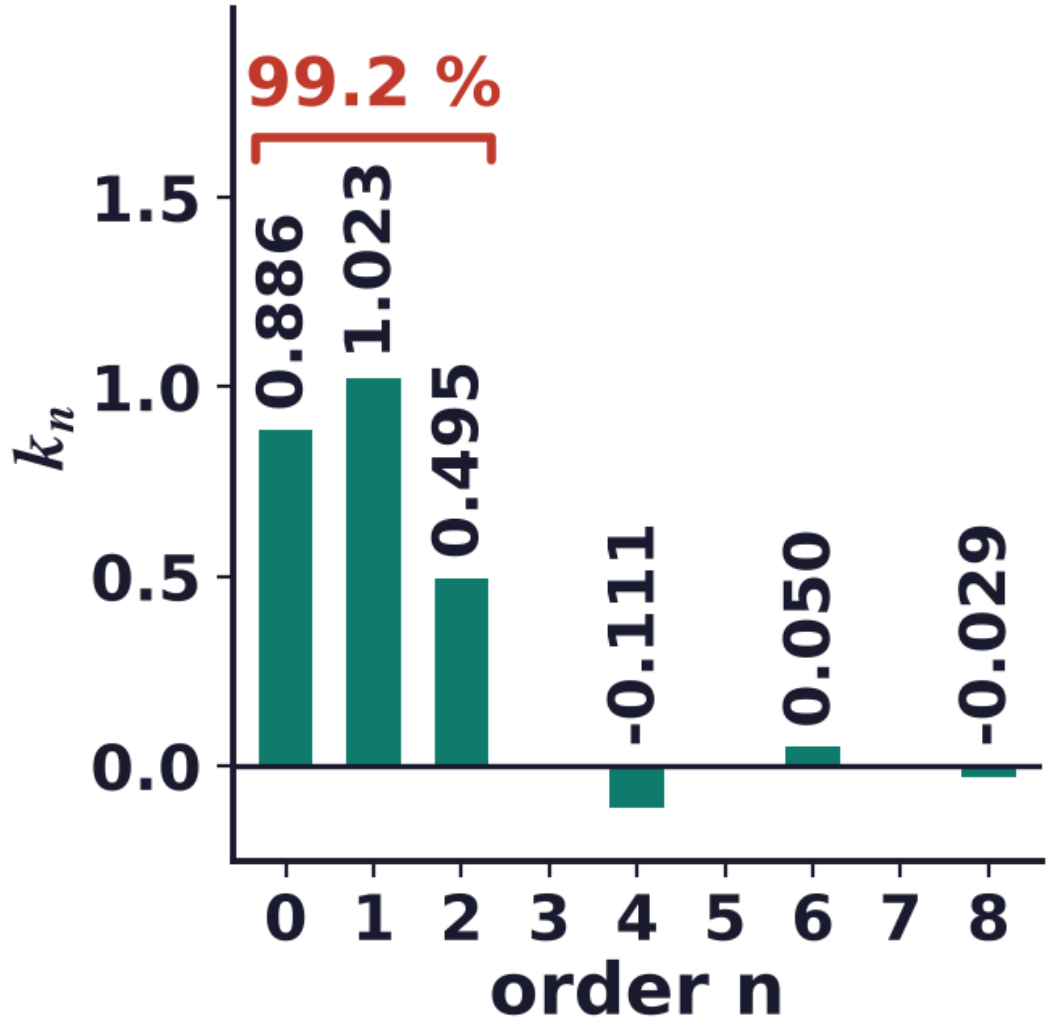



Figure 28: The coefficients $k_n$ of the kernel $\max(\cos\theta, 0)$ for $n = 0, \ldots, 8$ (teal); the bracket marks orders 0–2, with 99.2 % of the energy. One view sees only the normals facing the camera, so about 98 % is achieved with six basis images (Ramamoorthi, 2002).

# Harmonic models suit convex Lambertian objects

Harmonic representations handle convex Lambertian objects with multiple light sources, including attached shadows. They do not model cast shadows and inter-reflections, and specular objects may require a very large basis.

Their accuracy is maintained for fairly close light sources (Frolova et al., 2004). A point light near a sphere lights its near side more strongly than a distant light would, so the reflected light varies faster over the sphere and the low orders capture less of it. With orders up to 2 the accuracy exceeds 0.96 from about four radii; the lower orders approach their limits slowly.

The harmonic coefficients of a lighting may describe negative light. Non-negative light can be enforced by projecting the illumination cone onto the harmonic space: closed-form constraints exist for the first-order approximation, and higher orders use sampling or a Toeplitz matrix (Shirdhonkar and Jacobs, 2005).

# Harmonic models suit convex Lambertian objects

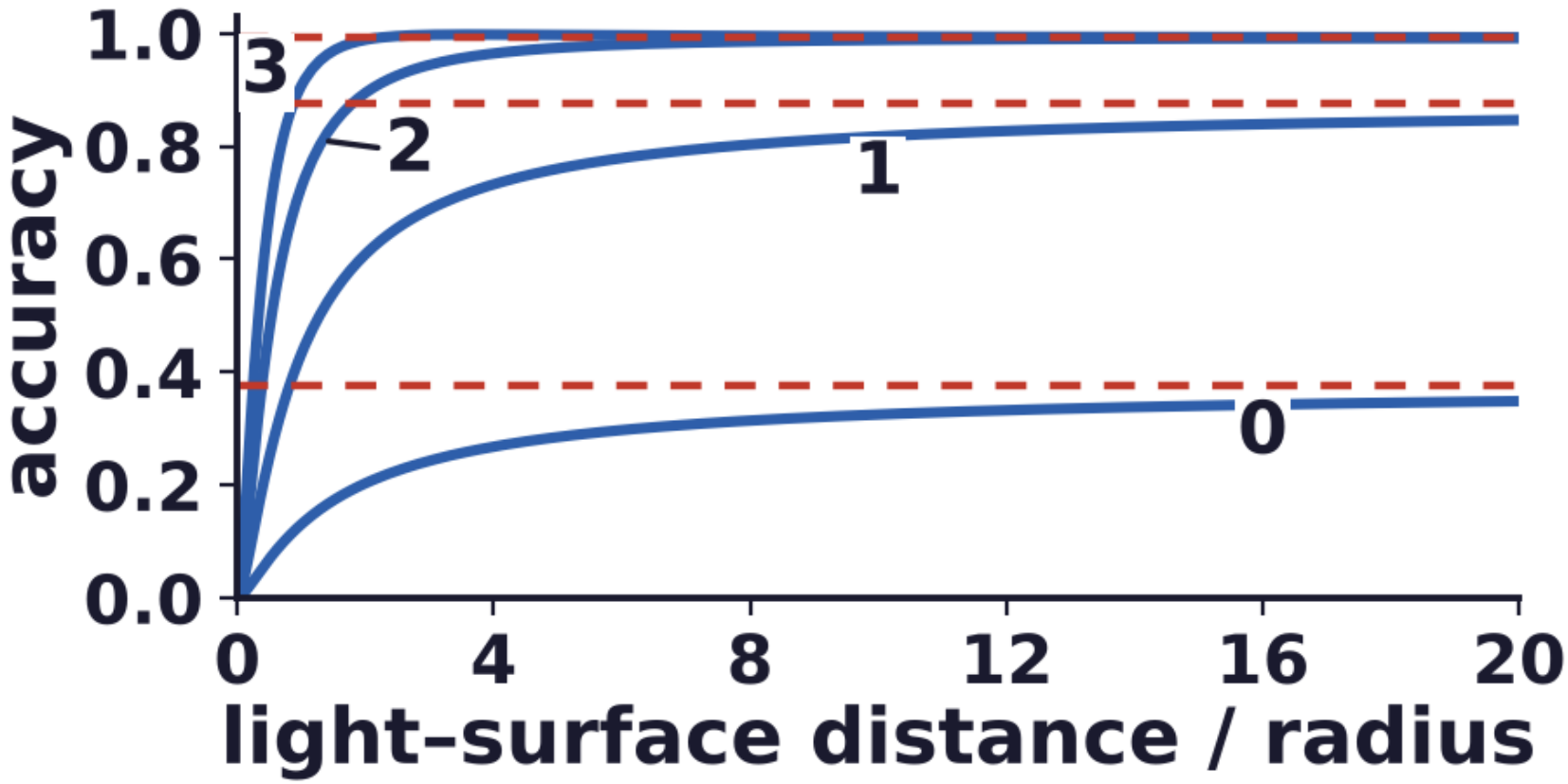



Figure 29: Accuracy of the harmonic approximations up to order 0, 1, 2 and 3 (blue) for a point light at a distance from the surface of a unit sphere, in radii; red dashed: the limits for a distant light, 0.375, 0.875 and 0.992 (the last for orders 2 and 3, since $k_3 = 0$). Computed after Frolova et al. (2004), B2 p. 13.

# Nine harmonic images approximate a face in any lighting

For an object with albedo $\rho(\vec{x})$ and normals $\vec{n}(\vec{x})$, the image under the lighting $\ell$ is approximately $\sum_{n \leq 2, m} \alpha_n l_{nm} \rho(\vec{x}) Y_{nm}(\vec{n}(\vec{x}))$: a combination of nine harmonic images $\rho\, Y_{nm}(\vec{n})$, each the albedo times a polynomial of the normal $(n_x, n_y, n_z)$, up to a constant:

$$\rho, \quad \rho n_z, \ \rho n_x, \ \rho n_y, \quad \rho(3n_z^2 - 1), \ \rho(n_x^2 - n_y^2), \ \rho n_x n_y, \ \rho n_x n_z, \ \rho n_y n_z.$$

(G writes $\rho(2n_x^2 - 1)$ for the fifth, which spans the same space; the zonal harmonic of order 2 is $3n_z^2 - 1$.) The images of a convex Lambertian object under all lightings lie close to the nine-dimensional space they span, so they can predict novel appearances under new lighting. Applications include face recognition, photometric stereo, 3D reconstruction with a prior and motion analysis.

# Nine harmonic images approximate a face in any lighting

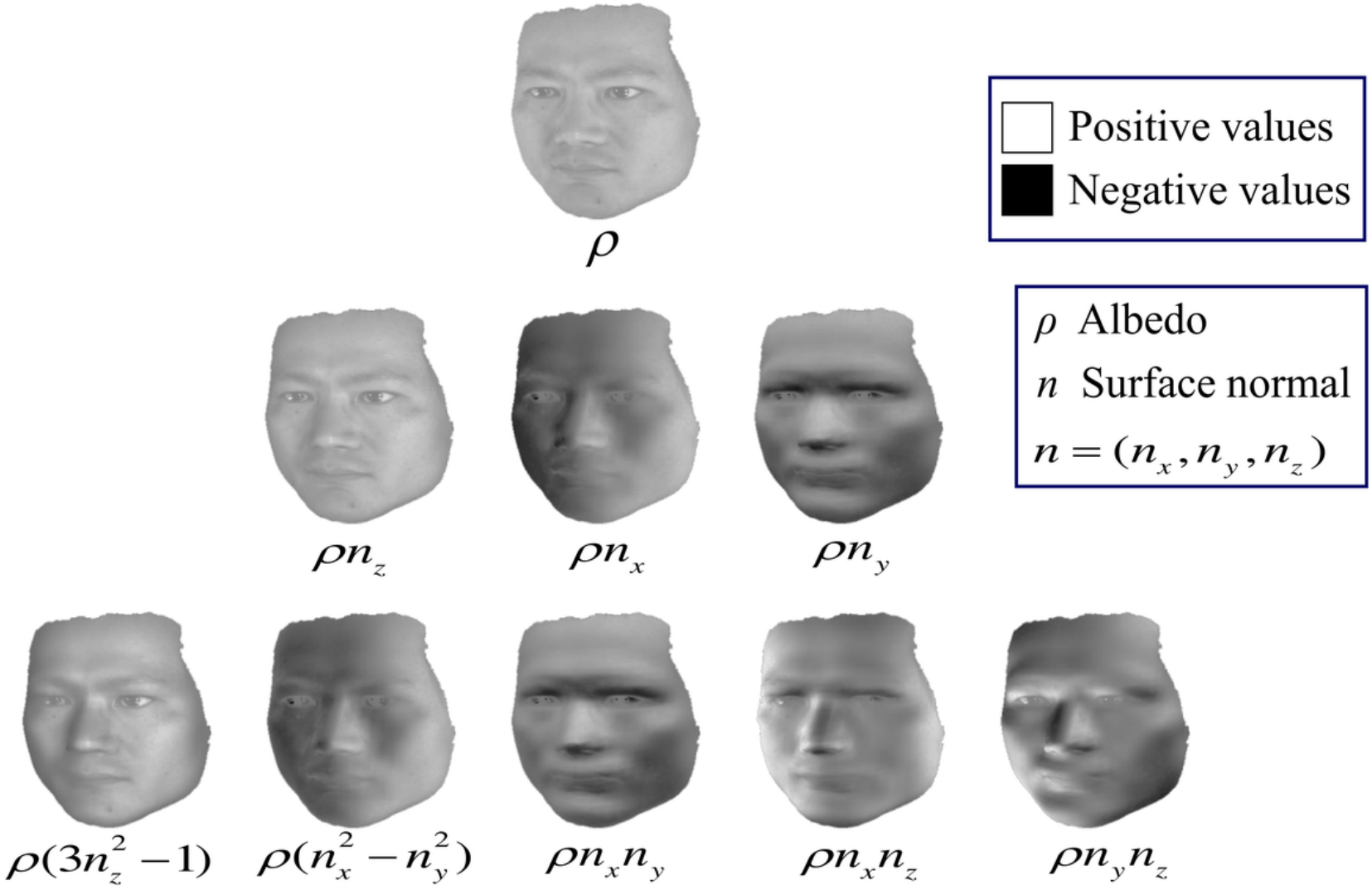



Figure 30: The nine harmonic images of a face, by order (white positive, black negative). B2 p. 15.

# Photometric stereo with harmonic images has rank nine

Photometric stereo works as before, with nine (or four) basis vectors in place of the three components of the scaled normal (Basri et al., 2007): $M \approx LS$, where $M$ ($f \times p$) holds the images, $L$ ($f \times 9$) the harmonic coefficients of each image's lighting, and $S$ ($9 \times p$) the harmonic images,

$$S = \big(\rho,\ \rho n_z,\ \rho n_x,\ \rho n_y,\ \rho(3n_z^2 - 1),\ \rho(n_x^2 - n_y^2),$$

$$\rho n_x n_y,\ \rho n_x n_z,\ \rho n_y n_z\big)^T.$$

The SVD of $M$ gives the best rank-$r$ factorization, $r = 9$ (or $r = 4$ with orders $\leq 1$), up to an $r \times r$ ambiguity, as for rank three. The lights may be general and unknown. Basri et al. reduce the ambiguity by requiring the rows of $S$ to have this structure (the first row $\rho$, the next three $\rho\vec{n}$ with $|\vec{n}| = 1$, the rest quadratic in $\vec{n}$), and integrability constrains the normals as before.

# Photometric stereo with harmonic images has rank nine

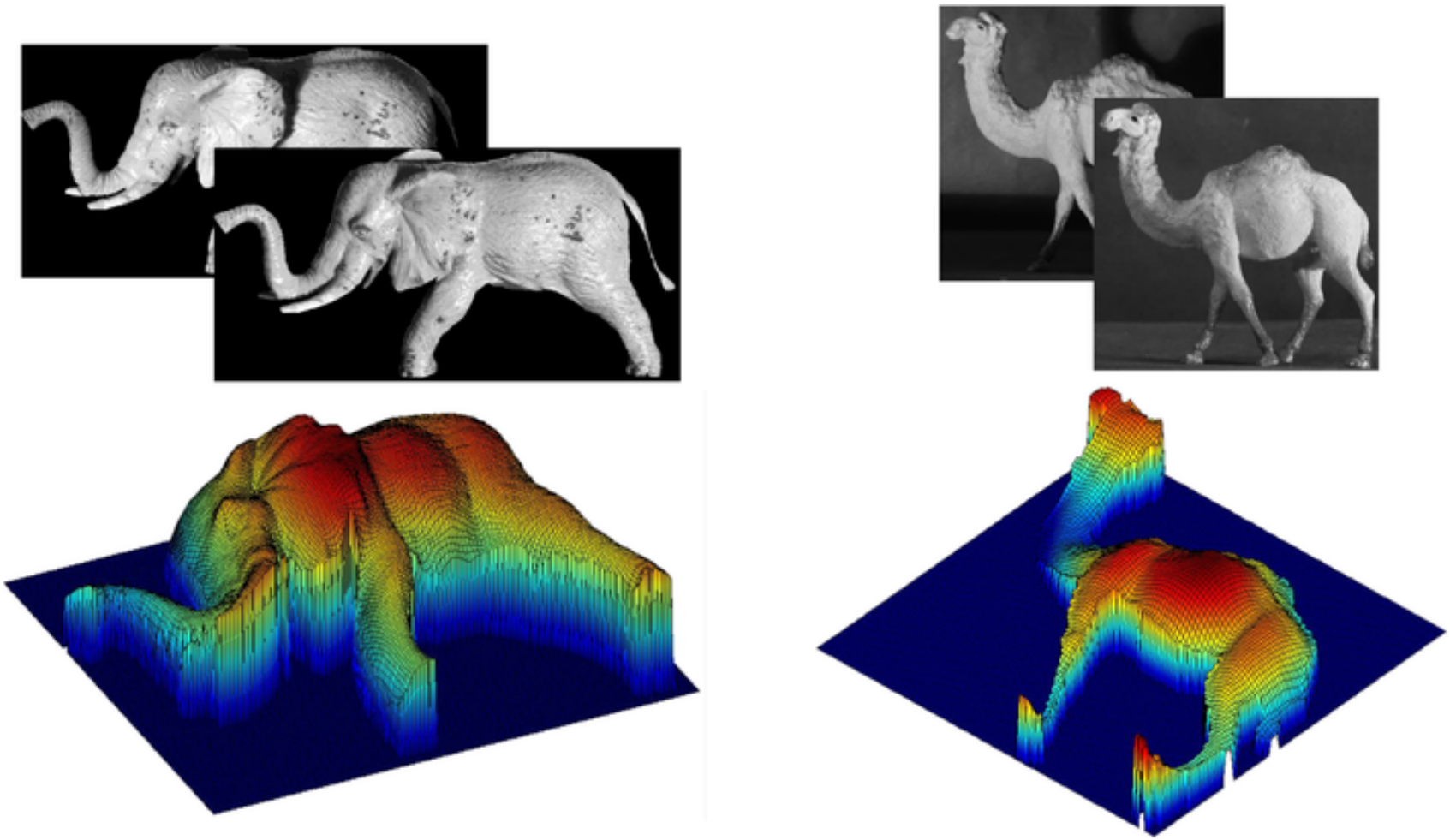


Figure 31: Images of an elephant and of a camel figurine, and the recovered depth (color is height). B2 p. 18.

# A reference face allows reconstruction from one image

Given just one image, shape from shading is impractical, but reconstruction is possible with a prior (Kemelmacher-Shlizerman and Basri, 2011). The prior is one reference face, its depth $Z_{prior}$ and albedo $\rho_{prior}$. The energy

$$\min_{\vec{l},\rho,Z} \iint_{\Omega} D + S, \qquad D = \left(I - \rho \vec{l}^{\,T} Y(\hat{n})\right)^2,$$

$$S = \lambda_1 \left(\Delta(Z - Z_{prior})\right)^2 + \lambda_2 \left(\Delta(\rho - \rho_{prior})\right)^2$$

sums over the image domain $\Omega$ a data term $D$ and a prior term $S$ (not the SVD's $D$ or the lights $S$). $Y(\hat{n})$ is the vector of the nine harmonics at the normal (harmonic images without the albedo) and $\vec{l}$ the nine lighting coefficients, so $D$ compares the image with the harmonic model. $\Delta$ is the Laplacian $(\partial_x^2 + \partial_y^2)$, and $S$ asks the depth and the albedo to differ from the reference's by smooth functions; $\lambda_1, \lambda_2$ weight the two terms. B2 solves it as a linear PDE; the paper works in steps: the lighting $\vec{l}$ by least squares with the reference's normals, then the depth $Z$ (linear once the normal's length is taken from the reference), then the albedo $\rho$.

# A reference face allows reconstruction from one image



Figure 32: Each photograph is followed by its recovered shape (two views for the last), then the shape with its recovered albedo; the red dashed box shows the reference face. B2 p. 20.

# Graphics renders images from 3D models; vision inverts it

Rendering, or image synthesis, generates an image from a 3D model by a computer program. Classic surface models describe how light is reflected off surfaces; Lambert's law dates back to the 18th century. CAD (computer-aided design) models represent an object by a polygon mesh, a collection of vertices, edges and faces, most simply triangles.

Graphics knows the objects and the light sources and computes the image; vision addresses the harder inverse problem of finding the objects and the lights from the image. Lambertian models are comparatively easy to invert, mainly because they are linear (if shadows are ignored) and the image does not depend on the viewpoint. Radiosity models (Goral et al., 1984) follow light bouncing between diffuse surfaces and give more realistic images, but they are hard to invert, as are BRDF and specularity models.

# Graphics renders images from 3D models; vision inverts it

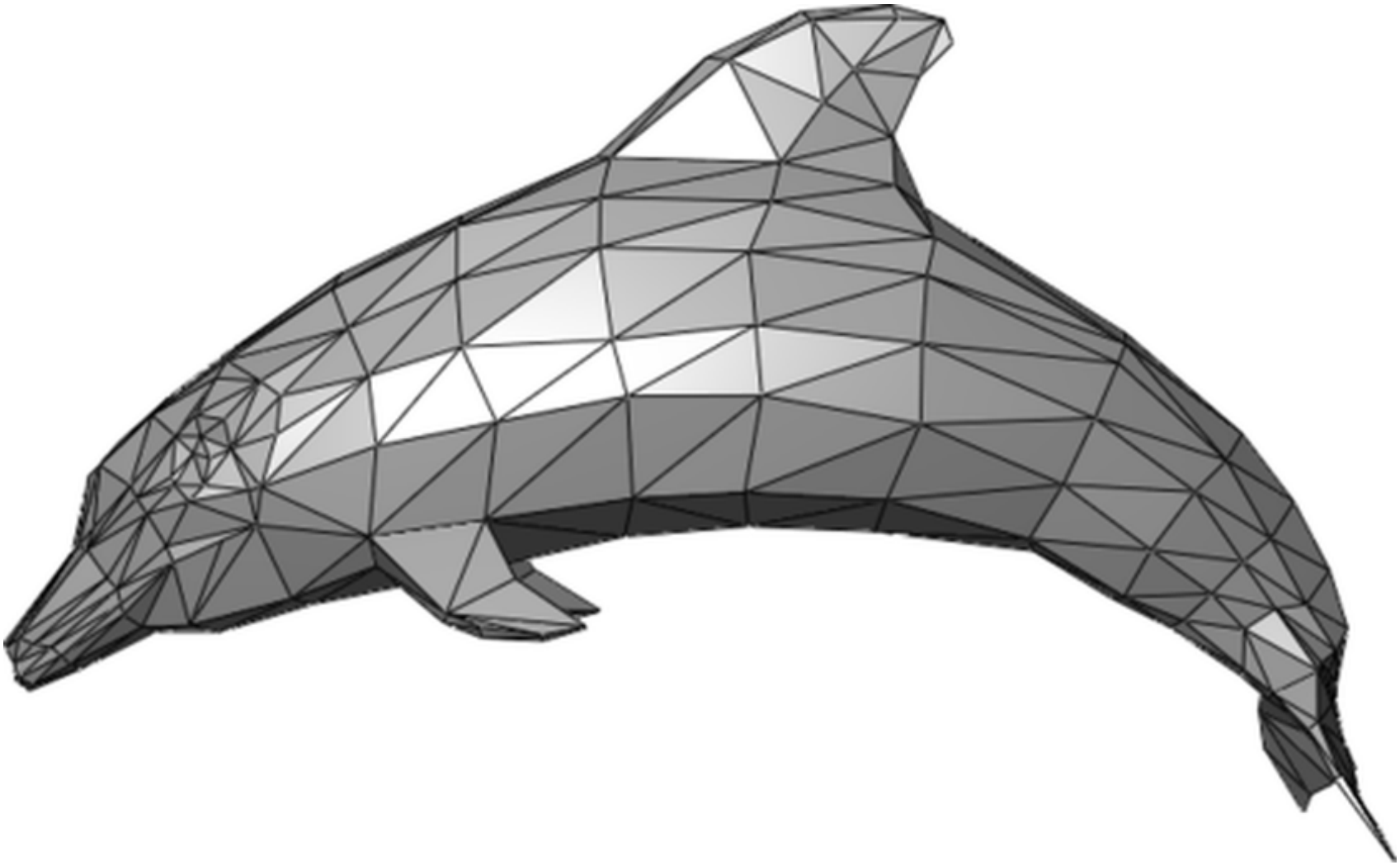


Figure 33: A polygon mesh of triangles. AN p. 5.

# Rasterization projects triangles; ray tracing casts rays

Rasterization projects each triangle of the mesh onto the image and determines the visible triangle at each pixel; a depth buffer keeps, at each pixel, the nearest triangle found so far. Properties stored at the vertices (colors, normals, texture coordinates) are interpolated inside the triangle. Ray tracing goes the other way: it casts a viewing ray through each pixel into the scene and traces the primitives that the ray meets.

For a pinhole camera at $\vec{o}$ with rotation $R$, focal length $f$ and principal point $(p_x, p_y)$ (the pixel where the optical axis meets the image), the ray through pixel $(u, v)$ is

$$\vec{r}(t) = \vec{o} + t\,\vec{d} \quad (t \geq 0), \qquad \vec{d} = R^{-1} \begin{pmatrix} (u - p_x)/f \\ (v - p_y)/f \\ 1 \end{pmatrix}.$$

The vector in parentheses is the pixel's direction in camera coordinates (unit depth); $R^{-1} = R^T$ turns it into world coordinates, and $t$ runs along the ray. In the convention of Lecture 16, $X' = R(X - \vec{t}\,)$, the camera location is $\vec{o} = \vec{t}$, the translation (a vector, unlike the ray parameter $t$).

# Rasterization projects triangles; ray tracing casts rays

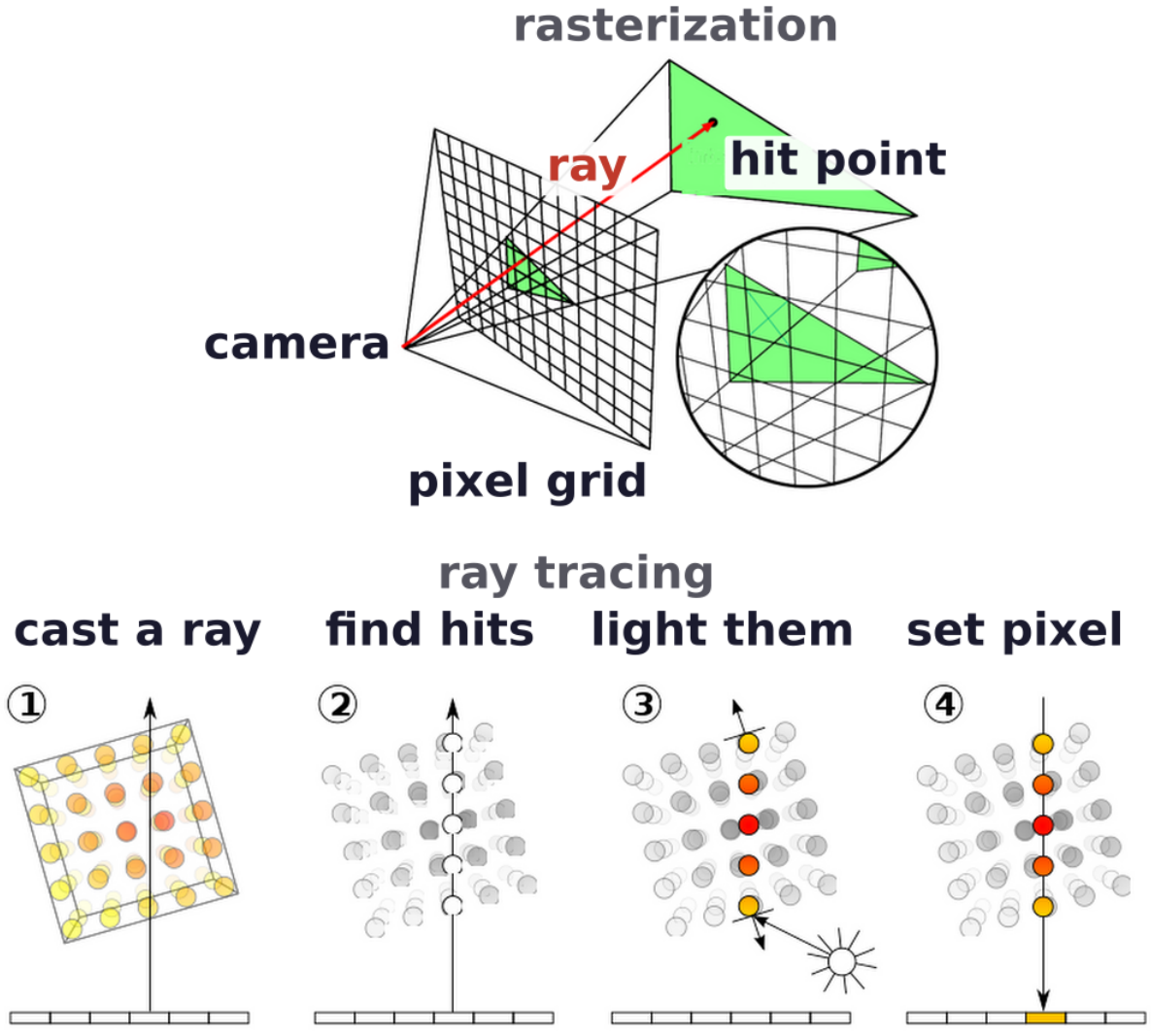



Figure 34: Top (rasterization): the triangle is projected onto the pixel grid (inset: the pixels it covers); the ray through one pixel shows the point of the triangle it sees (picture: www.scratchapixel.com, AN p. 6). Bottom: ray tracing in four steps (AN p. 11).

# Phong shading adds ambient, diffuse and specular terms

Phong (1975) lights each surface point by three terms:

$$I_p = k_a i_a + \sum_m \left( k_d \max(\hat{L}_m \cdot \hat{N}, 0)\, i_{m,d} + k_s \max(\hat{R}_m \cdot \hat{V}, 0)^\alpha\, i_{m,s} \right).$$

$I_p$ is the intensity at the surface point $p$; $m$ runs over the lights; $\hat{L}_m$ is the direction to light $m$, $\hat{N}$ the normal, $\hat{R}_m = 2(\hat{L}_m \cdot \hat{N})\hat{N} - \hat{L}_m$ the reflected ray and $\hat{V}$ the direction to the viewer. $k_a, k_d, k_s$ are the ambient, diffuse and specular reflection constants of the material, and $i_a, i_{m,d}, i_{m,s}$ the intensities of the ambient light and of the diffuse and specular components of light $m$. The ambient term is a constant that stands in for all indirect light; the diffuse term is Lambert's law; the specular term is a highlight around the mirror direction, narrower for a larger shininess $\alpha$. In the pipeline, the mesh is rasterized, the normal is found at each surface point, and the point is lit.

# Phong shading adds ambient, diffuse and specular terms

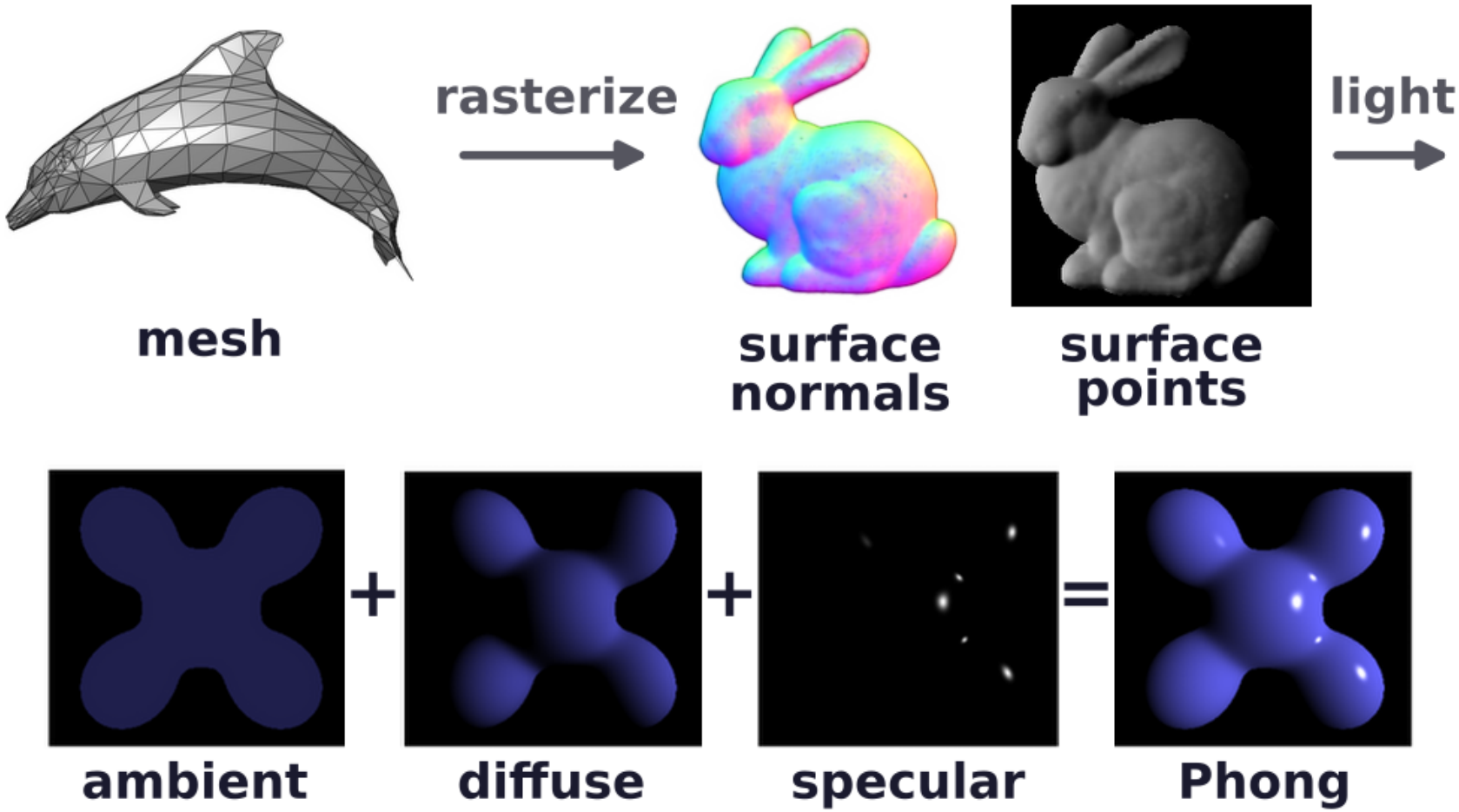



Figure 35: Top: a mesh is rasterized into surface normals (colored by direction) and surface points, which are then lit. Bottom: ambient + diffuse + specular = Phong, on an example shape. AN pp. 8–9.

# Differentiable rendering relates pixels back to 3D properties

Differentiable rendering writes the image as a function $I(\text{pixel}) = R(\text{vertices, faces, texture, camera})$ that can be differentiated, so image pixels can be related back to 3D properties of the scene (Ravi et al., 2020). Render, compare with the real image by a loss, and follow the gradient of the loss to change the vertices, the texture, the lights and the camera.

Hard coverage, whether a pixel lies inside a triangle, is a step function: its gradient is zero almost everywhere. The soft rasterizer (Liu et al., 2019) softens the edges: a triangle covers a pixel with probability $D = \mathrm{sigmoid}(\delta\, d^2/\sigma)$, where $d$ is the pixel's distance to the triangle's nearest edge, $\delta = +1$ inside and $-1$ outside, and $\sigma$ sets the width of the blur. Overlapping triangles are blended by depth, through a softmax over their depths weighted by $D$. Classic models (Lambertian, Phong, BRDF) suit single objects, but their parameters are set by hand; for scenes, complex lighting is probably best dealt with by learning methods.

# Differentiable rendering relates pixels back to 3D properties

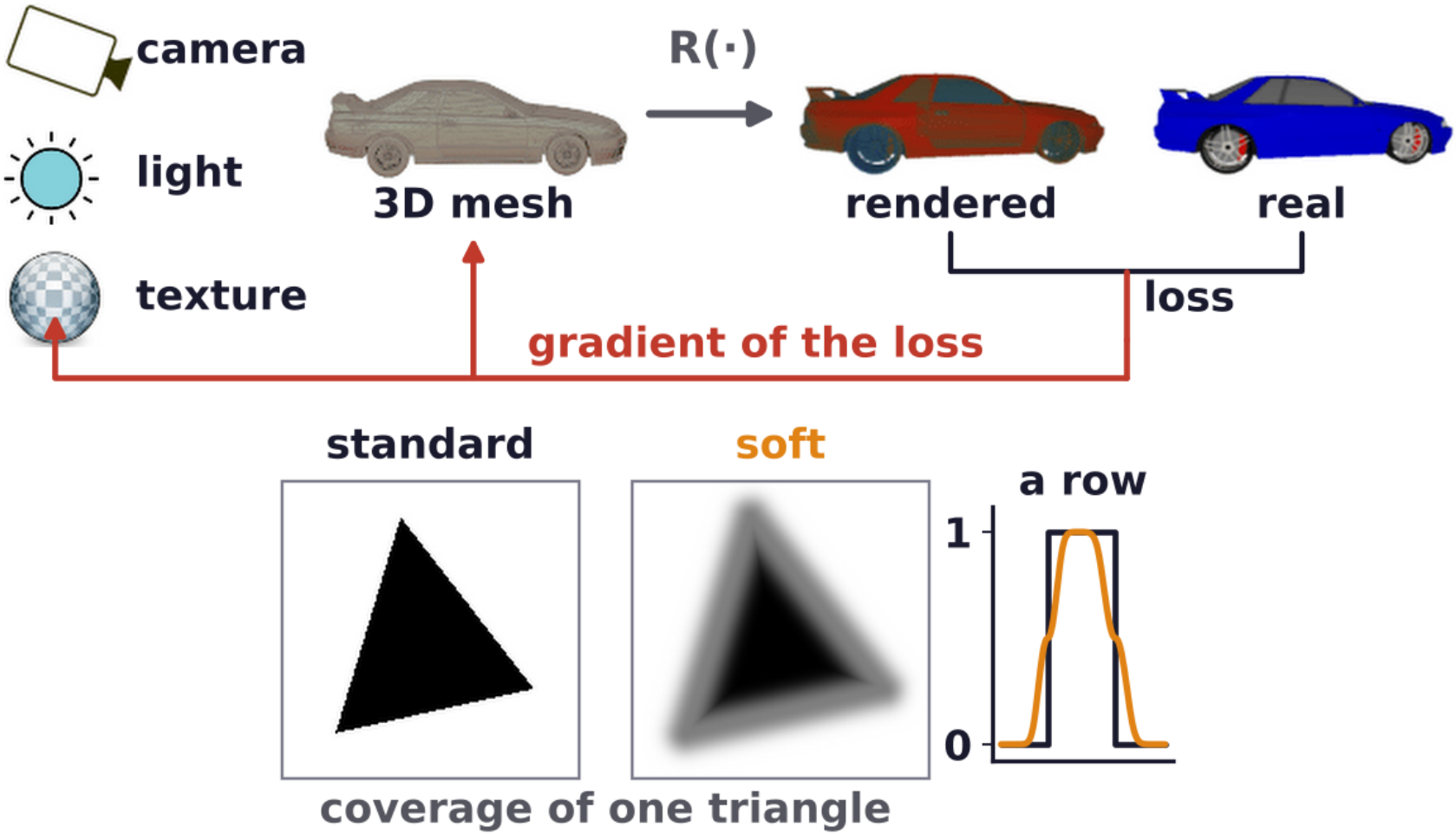



Figure 36: Top: camera, light, texture and a 3D mesh are rendered by $R$; the loss compares the result with the real image, and its gradient (red) flows back (after AN p. 4). Bottom: standard and soft coverage of one triangle, and one row of pixels across it.

# Summary

- Smooth surfaces reflect specularly and rough ones diffusely; the BRDF describes both.
- Lambertian model: $I(\vec{x}) = a(\vec{x})\,\vec{n}(\vec{x}) \cdot \vec{s}$, linear in the light and independent of the viewpoint.
- Attached and cast shadows break the linearity; a convex shape and a concave one can look alike.
- Photometric stereo: images under several lights have rank three; the SVD finds shape and lights.
- Shape and lights are found up to a $3 \times 3$ matrix; integrability leaves the generalized bas-relief ambiguity.
- Shape from shading (known light, constant albedo) solves an eikonal equation by fast marching.
- Lambertian reflectance smooths the lighting; nine harmonics capture at least 97.96 % of the reflectance.
- Rendering rasterizes or ray traces a mesh and shades it; differentiable rendering fits it by gradients.

# References


[1] A. L. Yuille. Lambertian models of reflectance. Lecture notes, Vision as Bayesian Inference, Johns Hopkins University, 2020 (G).

[2] R. Basri. Lambertian model of reflectance I: shape from shading and photometric stereo. Lecture slides (PowerPoint version, 38 slides), Weizmann Institute of Science (B1).

[3] R. Basri. Lambertian model of reflectance II: harmonic analysis. Lecture slides (PowerPoint version, 28 slides), Weizmann Institute of Science (B2).

[4] A. Wang and A. L. Yuille. Rendering technicals for differentiable rendering and implicit functions. Lecture slides, Johns Hopkins University (AN).

[5] Y. Adini, Y. Moses and S. Ullman. Face recognition: the problem of compensating for changes in illumination direction. *IEEE TPAMI*, 19(7):721–732, 1997.

[6] J. H. Lambert. *Photometria, sive de mensura et gradibus luminis, colorum et umbrae*. Eberhard Klett, Augsburg, 1760.

[7] R. L. Gregory. Knowledge in perception and illusion. *Phil. Trans. R. Soc. Lond. B*, 352(1358):1121–1127, 1997.

# References


[8] R. J. Woodham. Photometric method for determining surface orientation from multiple images. *Optical Engineering*, 19(1):139–144, 1980.

[9] A. L. Yuille, D. Snow, R. Epstein and P. N. Belhumeur. Determining generative models of objects under varying illumination: shape and albedo from multiple images using SVD and integrability. *IJCV*, 35(3):203–222, 1999.

[10] A. Shashua. On photometric issues in 3D visual recognition from a single 2D image. *IJCV*, 21(1–2):99–122, 1997.

[11] H. Hayakawa. Photometric stereo under a light source with arbitrary motion. *JOSA A*, 11(11):3079–3089, 1994.

[12] P. N. Belhumeur, D. J. Kriegman and A. L. Yuille. The bas-relief ambiguity. *IJCV*, 35(1):33–44, 1999.

[13] J. J. Koenderink, A. J. van Doorn, A. M. L. Kappers and J. T. Todd. Ambiguity and the 'mental eye' in pictorial relief. *Perception*, 30(4):431–448, 2001.

[14] A. L. Yuille, J. M. Coughlan and S. Konishi. The KGBR viewpoint-lighting ambiguity. *JOSA A*, 20(1):24–31, 2003.

# References


[15] B. K. P. Horn. Obtaining shape from shading information. In P. H. Winston (ed.), *The Psychology of Computer Vision*, 115–155. McGraw-Hill, New York, 1975.

[16] J. N. Tsitsiklis. Efficient algorithms for globally optimal trajectories. *IEEE Trans. Automatic Control*, 40(9):1528–1538, 1995.

[17] J. A. Sethian. A fast marching level set method for monotonically advancing fronts. *PNAS*, 93(4):1591–1595, 1996.

[18] R. Kimmel and J. A. Sethian. Optimal algorithm for shape from shading and path planning. *J. Mathematical Imaging and Vision*, 14(3):237–244, 2001.

[19] R. Zhang, P.-S. Tsai, J. E. Cryer and M. Shah. Shape from shading: a survey. *IEEE TPAMI*, 21(8):690–706, 1999.

[20] J. Oliensis. Uniqueness in shape from shading. *IJCV*, 6(2):75–104, 1991.

[21] J. J. Atick, P. A. Griffin and A. N. Redlich. Statistical approach to shape from shading: reconstruction of three-dimensional face surfaces from single two-dimensional images. *Neural Computation*, 8(6):1321–1340, 1996.

# References


[22] B. Potetz and T. S. Lee. Statistical correlations between two-dimensional images and three-dimensional structures in natural scenes. *JOSA A*, 20(7):1292–1303, 2003.

[23] P. N. Belhumeur and D. J. Kriegman. What is the set of images of an object under all possible illumination conditions? *IJCV*, 28(3):245–260, 1998.

[24] R. Ramamoorthi. Analytic PCA construction for theoretical analysis of lighting variability in images of a Lambertian object. *IEEE TPAMI*, 24(10):1322–1333, 2002.

[25] R. Basri and D. W. Jacobs. Lambertian reflectance and linear subspaces. *IEEE TPAMI*, 25(2):218–233, 2003.

[26] R. Ramamoorthi and P. Hanrahan. On the relationship between radiance and irradiance: determining the illumination from images of a convex Lambertian object. *JOSA A*, 18(10):2448–2459, 2001.

[27] D. Frolova, D. Simakov and R. Basri. Accuracy of spherical harmonic approximations for images of Lambertian objects under far and near lighting. *ECCV*, LNCS 3021, 574–587, 2004.

[28] S. Shirdhonkar and D. W. Jacobs. Non-negative lighting and specular object recognition. *ICCV*, 1323–1330, 2005.

# References


[29] R. Basri, D. Jacobs and I. Kemelmacher. Photometric stereo with general, unknown lighting. *IJCV*, 72(3):239–257, 2007.

[30] I. Kemelmacher-Shlizerman and R. Basri. 3D face reconstruction from a single image using a single reference face shape. *IEEE TPAMI*, 33(2):394–405, 2011.

[31] C. M. Goral, K. E. Torrance, D. P. Greenberg and B. Battaile. Modeling the interaction of light between diffuse surfaces. *SIGGRAPH, Computer Graphics*, 18(3):213–222, 1984.

[32] B. T. Phong. Illumination for computer generated pictures. *Communications of the ACM*, 18(6):311–317, 1975.

[33] N. Ravi, J. Reizenstein, D. Novotny, T. Gordon, W.-Y. Lo, J. Johnson and G. Gkioxari. Accelerating 3D deep learning with PyTorch3D. arXiv:2007.08501, 2020.

[34] S. Liu, T. Li, W. Chen and H. Li. Soft rasterizer: a differentiable renderer for image-based 3D reasoning. *ICCV*, 7708–7717, 2019.